

# Demand Shocks, Timing Preferences and the Equilibrium Term-Structures

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## ABSTRACT

Leading asset pricing models fail to describe the downward-sloping term-structure of equity. This paper proposes a closed-form framework where both supply- and demand-shocks contribute to equilibrium prices under either full or incomplete information. The joint role of timing preferences and demand-shocks on the equilibrium term-structures of bond and equity is characterized. The model reconciles standard asset pricing facts, such as the low and smooth risk-free rate, the high equity premium and volatility, with positive and negative slopes respectively of bond and equity. Under incomplete information, the minimum requirement on information quality, which preserves the slopes of the term-structures, is derived. Information quality barely affects the unconditional moments of equity returns but is crucial to their term-structure properties, which appear as an important diagnostics of asset pricing frameworks.

**Keywords.** stochastic differential utility · affine models · demand shocks · timing preferences · term structure of equity

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# I. Introduction

Standard asset pricing models fail to describe the main properties of financial markets mostly because they imply a strong connection among macroeconomic fundamentals and stocks returns, whereas asset prices fluctuations are barely uncorrelated with consumption in the real data. As introduced by [Cochrane and Hansen \(1992\)](#), such a “correlation puzzle” lies at the heart of all asset pricing puzzles, such as those concerning the risk-free rate, the equity premium and the excess return volatility.

A representative agent model featuring recursive preferences of the type considered by [Kreps and Porteus \(1979\)](#), [Epstein and Zin \(1989\)](#) and [Weil \(1989\)](#) partially solves the correlation puzzle. Indeed, in presence of latent factors affecting the conditional distribution of consumption, under recursive preferences, in addition to the transient consumption risk, also all variables which affect the future evolution of the agent’s utility deserve a compensation in equilibrium. [Bansal and Yaron \(2004\)](#) shows that small fluctuations in expected consumption growth can lead to a sizeable equity premium, a compensation for the so-called long-run risk.

However, long-run risk models seem to be inconsistent with the empirical evidence about the equilibrium term-structures of premia on bond and equity returns. Indeed, these models generate high levels for the premia at the cost of a wrong distribution of these premia over time. These models imply downward and upward sloping term-structures respectively for bond and equity returns, whereas the opposite slopes are observed in the real data. It is well known that bonds feature a positive term premium, whereas recently [van Binsbergen, Brandt, and Koijen \(2012\)](#) and [van Binsbergen, Hueskes, Koijen, and Vrugt \(2013\)](#) have documented the negative slope of the term-structure of equity, as implied by dividend strips.

The correlation puzzle arises since all uncertainty is loaded onto the supply side of the economy. In spirit of [Albuquerque, Eichenbaum, and Rebelo \(2013\)](#), I argue that instead demand shocks play a central role and help to reconcile the standard asset pricing facts with the new evidence about the term-structures. Demand shocks are introduced in the model in a very simple way. Namely, the state of the economy is characterized by a state-variable, which affects both the total resources –the usual supply side effect– as well as the perception of utility from consumption –the demand-side effect. The latter can be interpreted as, for instance, changes in the investors’ attitude towards saving and leisure, taste shocks induced by sentiment fluctuations, changes in the institutional and political

environment, or changes in the demographics of either the financial markets or the whole economy. Hence, in the model demand shocks represent the source of valuation risk associated to future cash-flows due to the perception of utility from consumption, conditional on the state of the economy.

In particular, the model is intended to capture, in the context of an endowment economy, the usual mutual feedback effect that we expect between supply- and demand-side effects in the context of a more general economy. Therefore, demand shocks are included in the model in such a way that the sensitivity of expected consumption, as perceived by the agent, with respect to the state of the economy is smaller than the sensitivity of actual consumption. As an example consider the case of a new technology: on the one hand, it can lead to an increase in productivity –the supply-side effect; on the other hand, it can induce a higher standard of living due to the use of the product of the new technology –the demand-side effect. As a result, supply and demand shocks can partially offset each other as measured in terms of the agent’s utility.

This paper proposes a closed-form and parsimonious asset pricing model to investigate how the timing preferences due to recursive utility and the demand shocks interact and affect both the equilibrium state-price process and the term-structures of bond and equity premia, under either full or incomplete information about the state of the economy.

The model works as follows. The representative agent of an endowment economy features stochastic differential utility in which the perceived consumption process differs from actual consumption. Namely, a state-variable process drives the conditional distribution of both actual consumption as well as perceived consumption. For the sake of comparison with [Bansal and Yaron \(2004\)](#), the state-variable process is a single mean-reverting process which induces fluctuations in expected consumption growth. In order to model in the context of an endowment economy –that is, in reduced form– the equilibrium mutual feedback between supply- and demand-side effects of a more general economy, I restrict the attention to the following case. The semi-elasticity of the expected perceived consumption with respect to the state-variable process at any future horizon is smaller than the semi-elasticity of actual consumption. Such an effect is modelled in a parsimonious way through a single parameter and in order: i) to preserve closed-form solutions to all equilibrium outcomes and ii) to let unchanged the timing preferences implied by the stochastic differential utility. Indeed, the agent’s preference for the timing resolution of uncertainty does not depend on the perceived consumption. As in [Skiadas \(1997, 1998\)](#), I say that the agent has preference for the early (late) resolution of uncertainty if the

aggregator function is convex (concave) in the utility argument.

Therefore, the model allows to analytically characterize the joint role of timing preferences and demand shocks on the equilibrium term structures of premia on both bond and equity returns. The analysis is conducted under either full information or incomplete information. In the latter case the state-variable process is not directly observable and, hence, allows to complement the analysis by considering the role of information quality and its interaction with both timing preferences and demand shocks.

The model leads to four main results. First, under full information and in absence of demand shocks, the timing preferences fully characterize the sign of the slopes of the equilibrium term-structures. The correct specification of the term-structures of premia only obtain if the agent has preference for the late resolution of uncertainty, that is when the model cannot generate a sizeable equity premium. Instead, in presence of demand shocks, the sign of the slopes of the equilibrium term-structures depends on the interaction of the timing preferences as well as the strength of demand shocks. If the latter are large enough, the correct specification of the term-structures can obtain under the early resolution of uncertainty, that is when the model can generate a sizeable equity premium.

Second, under the infinite horizon the risk-free rate process does not depends on the presence of demand shocks. Hence, its smooth dynamics can obtain in equilibrium in accord with the smooth dynamics of macroeconomics fundamentals. Instead, the prices of (non-transient) risk are affected by the demand shocks and, hence, can lead to high volatile stock returns and high equilibrium compensations.

Third, under incomplete information the agent dynamically learns about the state-variable process by observing a noisy signal. Therefore, the equilibrium term-structures are characterized taking into consideration the filtration generated by the information available to the agent. I define information quality as the inverse of the volatility of the signal and I investigate whether the degree of information quality can alter the sign of the slopes of the term-structures. In absence of demand shocks, information quality affects the sign of the slopes only under preferences for the late resolution of uncertainty, that is in the case of little empirical interest. Instead, in presence of demand shocks information quality becomes relevant in the case of empirical interest, that is under preferences for the early resolution of uncertainty and when demand shocks are large enough to produce the correct specification of the term-structures. Namely, I show that there exists a threshold representing the minimum degree of information quality necessary to

preserve the right sign of the slopes of the term-structures of premia on bond and equity returns.

Fourth, a simple numerical illustration shows that, even if very parsimonious, the model can match many empirical features of the financial markets. Namely, the model consists of five cash-flows parameters and three preference parameters. Under a reasonable risk aversion and demand shocks large enough, the model simultaneously captures i) the smooth dynamics of consumption; ii) the low and persistent variation of expected growth; iii) the excess of volatility of dividends over consumption; iv) the low and smooth risk-free rate; v) the high equity premium and the high volatility of stock returns; vi) the positive term premium on the discount bonds; vii) the upward sloping term-structure of premia on bond returns; viii) the downward sloping term-structure of premia on equity returns.

**Related literature.** A number of contributions relate the demand for assets to taste shocks:<sup>1</sup> this paper has a motivation similar to [Albuquerque, Eichenbaum, and Rebelo \(2013\)](#) but differs in two main respects. First, I link supply- and demand-side effects to a unique state-variable process and consider the effect of incomplete information, whereas [Albuquerque et al. \(2013\)](#) adds to the economy an exogenous time-preference shock under full information. Second, I analytically investigate the equilibrium term-structures of bonds and equity, whereas [Albuquerque et al. \(2013\)](#) focuses on the traditional asset pricing facts and cannot explain the negative slope of equity premia.

The paper relates with the literature on the stochastic differential utility, such as [Duffie and Epstein \(1992a,b\)](#) and [Skiadas \(1997, 1998\)](#). In particular, the paper is close to [Duffie, Schroder, and Skiadas \(1997\)](#) but differs in two respects. First, [Duffie et al. \(1997\)](#) analyses the term-structure of interest rates under incomplete information,<sup>2</sup> whereas I also study the term-structure of equity –which has recently attracted the attention of empirical research and poses a hard challenge to standard equilibrium models. Second, [Duffie et al. \(1997\)](#) does not consider the role of demand shocks and their interaction with timing preferences.

The term-structure of interest rates has been considered in a large literature, since [Vasicek \(1977\)](#), [Hull and White \(1990\)](#) and [Cochrane and Piazzesi \(2005\)](#) among many others. Few attempts have try to explain the recent evidence about the term-structure of equity. [Belo, Collin-Dufresne, and Goldstein \(2014\)](#), [Ai, Croce, Diercks, and Li \(2012\)](#), [Marfè \(2013, 2014\)](#) investigate the links with the real economic activity, whereas [Croce, Lettau, and Ludvigson](#)

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<sup>1</sup>See for example [Garber and King \(1983\)](#) and [Campbell \(1986\)](#), or more recently [Normandin and St-Amour \(1998\)](#), [Eggertsson and Woodford \(2004\)](#), [Pavlova and Rigobon \(2007\)](#) and [Marfè \(2010\)](#).

<sup>2</sup>Incomplete information under state additive preferences was already investigated in [Dothan and Feldman \(1986\)](#), [Detemple \(1986, 1991\)](#) and [Gennotte \(1986\)](#), among others.

(2007) and [Berrada, Detemple, and Rindisbacher \(2013\)](#) focus on non-standard beliefs and preferences. This paper contributes to this literature studying the role of demand shocks and providing a simple and alternative explanation of the empirical evidence. In particular, the model allows for a comparison with the equilibrium model of [Bansal and Yaron \(2004\)](#) and the long-run risk literature.

The paper is organized as follows. Section II introduces the model's assumptions. Section III and IV characterize the equilibrium and the security prices respectively under finite and infinite horizon and analyse the effects of timing preferences and demand shocks on the equilibrium term-structures. Section V introduces incomplete information and investigate the role of information quality. A numerical illustration and discussion of the model's quantitative predictions is provided in Section VI. Section VII concludes.

## II. Consumption and Preferences

Consider a filtered probability space  $(\Omega, \mathcal{F}, \mathbf{F}, \mathbb{P})$  over a finite or infinite time-horizon  $(0, T)$ . Let the filtration  $\mathbf{F}$  contains the augmented filtration generated by a  $n$ -dimensional Brownian motion  $\mathbf{B}$ .

There is a single agent in the economy who consumes an endowment  $y$ , adapted to  $\mathbf{F}$ , that satisfies<sup>3</sup>

$$\frac{dy_t}{y_t} = \mu_y(\mathbf{x}_t)dt + \sigma_y(\mathbf{x}_t)'d\mathbf{B}_t, \quad 0 \leq t \leq T. \quad (1)$$

Here,  $\mu_y(\cdot) : \mathbb{R}^m \rightarrow \mathbb{R}$  and  $\sigma_y(\cdot) : \mathbb{R}^m \rightarrow \mathbb{R}^n$  and the initial value  $y_0$  is assumed to be a given constant. The  $m$ -dimensional state variable process  $\mathbf{x}$  evolves according to

$$d\mathbf{x}_t = \mu_x(\mathbf{x}_t)dt + \sigma_x(\mathbf{x}_t)'d\mathbf{B}_t, \quad 0 \leq t \leq T, \quad (2)$$

where  $\mu_x(\cdot) : \mathbb{R}^m \rightarrow \mathbb{R}^m$  and  $\sigma_x(\cdot) : \mathbb{R}^m \rightarrow \mathbb{R}^{n \times m}$  and the initial value  $\mathbf{x}_0$  is a constant.

As discussed in the Introduction, the agent derives utility from consumption as subject to demand shocks and, potentially, from the information filtration. Namely, the agent cares not only about the distribution of the sample path

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<sup>3</sup>I assume that standard conditions for the existence and uniqueness of strong solutions to all of the stochastic differential equations hold, throughout the paper (see [Karatzas and Shreve \(1988\)](#)).

for consumption, but also about the manner in which the conditional distribution of the consumption path, perturbed by demand shocks, evolves over time.

Given any sub-filtration  $\mathbf{D} = \{\mathcal{D}_t : t \in (0, T)\}$  of  $\mathbf{F}$  and any measurable consumption process  $c = \{c_t : t \in (0, T)\}$  adapted to  $\mathbf{D}$ , with  $c = y$  in equilibrium, I define the *perceived* consumption process  $\tilde{c} = \{\tilde{c}_t : t \in (0, T)\}$  as the measurable process adapted to  $\mathbf{D}$  given by

$$\tilde{c}_t = \mathfrak{C}(y_t, \mathbf{x}_t), \quad 0 \leq t \leq T, \quad (3)$$

where  $\mathfrak{C}(\cdot, \cdot) : \mathbb{R} \times \mathbb{R}^m \rightarrow \mathbb{R}$  is the primitive function determining the effects of demand shocks on the agent's perception of consumption  $\tilde{c}$ . In order to capture in reduced form the feedback effect due to demand shocks discussed in the Introduction, I restrict  $\mathfrak{C}(\cdot, \cdot)$  as follows.

**Assumption 1** *The perceived consumption process  $\tilde{c}$  is given by  $\tilde{c}_t = \mathfrak{C}(y_t, \mathbf{x}_t)$  for any  $0 \leq t \leq T$ , with  $\mathfrak{C}(\cdot, \cdot)$  satisfying:*

$$\frac{\nabla_{\mathbf{x}} \mathbb{E}[\tilde{c}_{t+\tau}^a | \mathcal{D}_t]}{\mathbb{E}[\tilde{c}_{t+\tau}^a | \mathcal{D}_t]} - \frac{\nabla_{\mathbf{x}} \mathbb{E}[c_{t+\tau}^a | \mathcal{D}_t]}{\mathbb{E}[c_{t+\tau}^a | \mathcal{D}_t]} \leq 0, \quad \forall 0 \leq \tau \leq T - t,$$

for any  $a \geq 0$  such that  $\mathbb{E}[c_{t+\tau}^a | \mathcal{D}_t]$  exists finite.

Assumption 1 states that the semi-elasticity of the moment generating function of perceived (log) consumption with respect to the state process  $\mathbf{x}$  has to be lower than the semi-elasticity of the moment generating function of (log) consumption at any future horizon  $\tau$ . Therefore, Assumption 1 is intended to capture the general case in which the sensitivity of expected consumption, as perceived by the agent, with respect to the state process is lower than that of actual consumption.

The agent's utility process,  $V(\tilde{c}, \mathbf{D})$ , is defined under usual technical conditions as the unique solution of the backward integral equation:

$$V_t(\tilde{c}, \mathbf{D}) = \mathbb{E} \left[ \int_t^T f(\tilde{c}_u, V_u(\tilde{c}, \mathbf{D})) du \mid \mathcal{D}_t \right], \quad 0 \leq t \leq T, \quad (4)$$

where  $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$  is the primitive function determining preferences. The process  $V_t(\tilde{c}, \mathbf{D})$  should be interpreted

as the time  $t$  utility of the remaining perceived consumption and filtration, with agent's utility at time zero defined as  $U(\tilde{c}, \mathbf{D}) = V_0(\tilde{c}, \mathbf{D})$ . [Duffie and Epstein \(1992b\)](#) discuss the existence of a unique integrable solution to Eq. (4), under appropriate assumptions on  $f$ . Such a utility process is known as stochastic differential utility since it has been derived as a limiting case of the discrete-time recursive utility of [Epstein and Zin \(1989\)](#).

Let  $\mathbf{F}^{y,x} = \{\mathcal{F}_t^{y,x} : 0 \leq t \leq T\}$  be the filtration generated by the endowment process  $y$  and the state-variable process  $x$ . Under full information,  $\mathbf{F}^{y,x}$  represents the total information observed by the agent: in such a case I denote with  $V$  the unique solution to Eq. (4) with  $\tilde{c}$  as in Eq. (3) and  $\mathbf{D} = \mathbf{F}^{y,x}$ . In Section V, I consider the case of incomplete information and I let the total information observed by the agent as given by a filtration  $\mathbf{F}^o = \{\mathcal{F}_t^o : 0 \leq t \leq T\}$  defined later.

Here, I assume a specific parametric form of  $f$  and obtain an expression for both the utility processes  $V(\tilde{c}, \mathbf{F}^{y,x})$  and  $V(\tilde{c}, \mathbf{F}^o)$  in closed form, under a particular specification of Eq. (1)-(2)-(3). The specific parametric form of  $f$  adopted in the paper is a special case of that considered by [Duffie and Epstein \(1992a\)](#):

$$f(c, v) = \beta(1 + \alpha v) \left( \log(c) - \frac{\log(1 + \alpha v)}{\alpha} \right) \quad (5)$$

for some constants  $\alpha$  and  $\beta$  satisfying  $0 \neq \alpha < 1$  and  $\beta > 0$ . Eq. (5) implies that coefficient of relative risk aversion is given by  $\gamma = 1 - \alpha$  and the elasticity of intertemporal substitution  $\psi$  equals unity. Such a restriction on the preferences provides enough tractability to focus on the scope of the paper and analytically discuss the role of timing preferences. In particular, as it will be shown later, for  $\alpha < 0$  ( $\alpha > 0$ ) the aggregator  $f(c, v)$  is convex (concave) in the utility argument and, hence, the agent has preference for the early (late) resolution of uncertainty. Notice that as  $\alpha \rightarrow 0$ ,  $f(c, v) \rightarrow \beta(\log(c) - v)$ . Indeed, for  $\alpha = 0$ , the resulting utility process has additive form given by

$$U(\tilde{c}, \mathbf{D}) = \beta \mathbb{E} \left[ \int_0^T e^{-\beta u} \log(\tilde{c}_u) du \right], \quad \alpha = 0.$$

Hereafter, I specialize Eq. (1)-(2)-(3) as follows i) for the sake of comparability with existing literature, ii) in order to provide intuitive economic interpretability, and iii) for the sake of analytic tractability and exposition.

Exogenous uncertainty of Eq. (1)-(2) specializes to the continuous-time counterpart of the endowment process

considered in [Bansal and Yaron \(2004\)](#) and in most of long-run risk literature:

$$\frac{dy_t}{y_t} = (\mu + s_x x_t)dt + \sigma dB_{y,t}, \quad (6)$$

$$dx_t = \lambda(\bar{x} - x_t)dt + \eta dB_{x,t}, \quad (7)$$

with  $n = 2, m = 1$  and  $\mathbf{B} = (B_y, B_x)$  and  $\mu, \sigma, \bar{x}, \eta$  and  $s_x > 0, \lambda > 0$  are constants.<sup>4</sup> Therefore, the state-variable process  $x$  is a mean-reverting process which drives the time-variation in the expected growth of the endowment process. The function of Eq. (3), which captures the effect of demand shocks on the agent's perception of consumption, specializes to the simple form given by

$$\tilde{c}_t = \mathfrak{C}(y_t, x_t) = y_t \exp\left(-\frac{d_x}{\beta} x_t\right), \quad (8)$$

where  $d_x$  is a constant, which I label demand-effect as opposed to the supply-effect  $s_x$ . The sign of  $d_x$  says if the effect of demand shocks increases or decreases with expected growth: namely, the condition on  $\mathfrak{C}(\cdot, \cdot)$  in Assumption 1 is satisfied for any  $d_x \geq 0$ .<sup>5</sup> As it will be shown,  $\text{sign}(d_x - s_x)$  has, jointly with  $\alpha$ , a crucial role on the characterization of the properties of the equilibrium term-structures of both bonds and equity. Notice that the agent's preference for the resolution of uncertainty do not depend on the value of  $d_x$ , indeed, as it will be shown later, the sign of  $f_{v,v}(\tilde{c}, v)$  does not depend on  $d_x$ .

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<sup>4</sup>The restriction  $s_x > 0$  is imposed only to interpret expected growth  $\mathbb{E}[dy_t | \mathcal{F}_t^{y,x}]/(y_t dt)$  as increasing with the state-variable process  $x_t$ , whereas the restriction  $\lambda > 0$  induces mean-reversion towards  $\bar{x}$  to the dynamics of  $x_t$ .

<sup>5</sup>In particular, the condition in Assumption 1 reduces to

$$\frac{\nabla_{\mathbf{x}} \mathbb{E}[\tilde{c}_{t+\tau}^a | \mathcal{F}_t^{y,x}]}{\mathbb{E}[\tilde{c}_{t+\tau}^a | \mathcal{F}_t^{y,x}]} - \frac{\nabla_{\mathbf{x}} \mathbb{E}[c_{t+\tau}^a | \mathcal{F}_t^{y,x}]}{\mathbb{E}[c_{t+\tau}^a | \mathcal{F}_t^{y,x}]} = -e^{-\lambda\tau} (d_x/\beta) a \leq 0$$

for any  $d_x \geq 0$ . Hence, the difference among the two semi-elasticities is monotone increasing with the horizon  $\tau$ . See the Appendix for the details.

### III. Finite Horizon Economy

#### A. Value function and state-price density

The utility process  $V = V(\tilde{c}, \mathbf{F}^{y,x})$  can be computed in closed form, after characterizing it as a solution to a partial differential equation.

**Proposition 1** *The process  $V = V(\tilde{c}, \mathbf{F}^{y,x})$  is given by*

$$V_t = \frac{1}{\alpha} \left( e^{\alpha(q_t \log y_t + h_t x_t + k_t)} - 1 \right), \quad (9)$$

where  $q$ ,  $h$  and  $k$  are deterministic processes, given by

$$q_t = 1 - e^{-\beta(T-t)}, \quad (10)$$

$$h_t = \frac{1}{\lambda(\beta + \lambda)} \left( \lambda(s_x - d_x) - e^{-\beta(T-t)} s_x(\beta + \lambda) + e^{-(\beta + \lambda)(T-t)} (s_x \beta + d_x \lambda) \right), \quad (11)$$

$$k_t = \int_t^T e^{-\beta(u-t)} \left( \bar{x} \lambda h_u + \frac{\alpha}{2} \eta^2 h_u^2 - q_u (\sigma^2 (1 - \alpha q_u) / 2 - \mu) \right) du. \quad (12)$$

The utility process  $V$  is exponential affine in  $\log y_t$  and  $x_t$ . Namely,  $V$  as a function of  $y_t$  is monotonically increasing, and  $V$  as a function of  $x_t$  is increasing if  $s_x > 0$  and  $d_x \leq 0$  and non necessarily monotone for  $s_x > 0$  and  $d_x > 0$ .

Following the Markovian approach of [Duffie and Epstein \(1992a,b\)](#), the equilibrium state-price formula is given by

$$\pi_t = \exp \left( \int_0^t f_v(\tilde{c}_u, V_u) du \right) f_c(\tilde{c}_t, V_t),$$

where  $f_v$  and  $f_c$  denote the two partial derivatives of  $f$ .<sup>6</sup> An application of Itô's lemma leads to the following characterization of the dynamics of  $\pi$ .

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<sup>6</sup>Notice that  $f_c(\tilde{c}, V) = \partial_c f(\tilde{c}, V) = \beta(1 + \alpha V) \partial_c \log(\tilde{c}) = \beta(1 + \alpha V)/c$  depends on  $d_x$  only through  $V$  and, hence, only for  $\alpha \neq 0$ .

**Proposition 2** *The state-price process  $\pi$  satisfies*

$$\begin{aligned}\frac{d\pi_t}{\pi_t} &= \frac{df_c(\tilde{c}_t, V_t)}{f_c(\tilde{c}_t, V_t)} + f_v(\tilde{c}_t, V_t)dt \\ &= -r_t dt - \theta_{y,t} dB_{y,t} - \theta_{x,t} dB_{x,t}, \quad t \in (0, T),\end{aligned}\tag{13}$$

where

$$\begin{aligned}r_t &= \beta + \mu + s_x x_t - \sigma^2 \\ &\quad - \alpha \bar{\lambda} h_t - \alpha^2 \eta^2 h_t^2 / 2 + \beta \alpha k_t - \alpha k'_t - \alpha q_t (\mu - \sigma^2 3/2 - \alpha q_t \sigma^2 / 2),\end{aligned}\tag{14}$$

and

$$\theta_{y,t} = \sigma(1 - \alpha q_t),\tag{15}$$

$$\theta_{x,t} = -\alpha \eta h_t.\tag{16}$$

The process  $r$  represents the instantaneous interest rate process: Eq. (14) is linear in  $x_t$  and contains all the information relevant to estimating the state variable. Moreover, the instantaneous variance of  $r$  is proportional to that of  $x_t$ . The coefficient of  $x_t$  in Eq. (14) is  $s_x$  and, hence, does not depend on the demand-effect  $d_x$ .

The price of risk associated with the Brownian motion  $B_y$  is denoted by  $\theta_y$ : as usual, it is given by the instantaneous volatility of  $\log y_t$  times a term,  $1 - \alpha q_t$ , which accounts for both the timing preferences and the remaining time-horizon. The function  $q_t$  belongs to  $(0, 1)$  and decreases with the time-horizon  $T - t$ , converging to zero when  $t \rightarrow T$ . Hence, when the remaining time-horizon approaches to zero  $\theta_y$  reduces to the instantaneous volatility of  $\log y_t$ , despite the timing preferences. At shorter horizons, these matter: if the agent has preference for the late resolution of uncertainty ( $\alpha > 0$ ), then  $\theta_y < \sigma$ , whereas if the agent has preference for the early resolution of uncertainty ( $\alpha < 0$ ), then  $\theta_y > \sigma$ .

The price of risk associated with the Brownian motion  $B_x$  is denoted by  $\theta_x$ : it is a price for non-transient risk and, hence, it disappears if the utility reduces to the additive form ( $\alpha \rightarrow 0$ ). In absence of demand shocks ( $d_x = 0$ ),  $h_t$  is

positive and, hence,  $\theta_x$  is larger than zero if the agent has preference for the early resolution of uncertainty ( $\alpha < 0$ ) and vice-versa. Moreover,  $\theta_x$  increases in magnitude with the remaining time-horizon ( $\partial_{T-t} h_t > 0$ ) and approaches to zero when  $t \rightarrow T$ . Instead, when the demand-effect is positive ( $d_x > 0$ ) the sign of  $h_t$  and  $\partial_{T-t} h_t$  is not necessarily positive. Therefore, for  $d_x$  large enough,  $\theta_x$  can be positive for  $\alpha > 0$  and for some remaining time-horizons  $T - t$ .

## B. Equilibrium security prices

The equilibrium price process  $S$  of any security can be described by the expected discounted value of a cumulative dividend process  $\delta$  (adapted and of integrable variation), where the expectation is computed under the filtration of the agent,  $\mathbf{F}^{y,x}$ , and discounting is made by use of the state-price process,  $\pi$ :

$$S_t = \frac{1}{\pi_t} \mathbb{E} \left[ \int_0^\tau \pi_{t+u} \delta_{t+u} du \mid \mathcal{F}_t^{y,x} \right], \quad 0 \leq \tau \leq T - t.$$

In spirit of [Abel \(1999\)](#), I define the dividend process representative of the stock markets as a levered version of the endowment process:  $\delta = y^\rho$ , with  $\rho \in \mathbb{R}$ . For  $\rho > 1$ , the process  $\delta$  captures the usual excess of volatility of market dividends over aggregate consumption, which is observed in the real data.

Namely, I am interested in two types of assets which pay a unique payoff at a given maturity  $t + \tau$  for any  $0 \leq \tau \leq T - t$ :

$$\begin{aligned} Q_{t,\tau} &= \frac{1}{\pi_t} \mathbb{E} [\pi_{t+\tau} \mid \mathcal{F}_t^{y,x}], \\ P_{t,\tau} &= \frac{1}{\pi_t} \mathbb{E} [\pi_{t+\tau} y_{t+\tau}^\rho \mid \mathcal{F}_t^{y,x}]. \end{aligned}$$

The former is the price process of a discount bond that matures at time  $t + \tau$ , while the latter is the price process a so-called dividend strip or equity claim associated to the dividend process  $\delta = y^\rho$ . Therefore, we use both  $Q_{t,\tau}$  and  $P_{t,\tau}$  to study the term-structures of interest rates and equity returns.

**Proposition 3** *The equilibrium price of the discount bond with residual maturity  $\tau$  is given by:*

$$Q_{t,\tau} = e^{a_0(\tau) + a_x(\tau)x_t}, \quad 0 \leq \tau \leq T - t. \quad (17)$$

where

$$a_0(\tau) = \int_0^\tau (\beta + \mu - \sigma^2 - \frac{1}{2}\eta^2 a_x(u)^2 - a_x(u) (\bar{x}\lambda + \alpha\eta^2 h_u) - \frac{1}{2}\alpha (2\bar{x}\lambda h_u + \alpha\eta^2 h_u^2 - 2\beta k_u + q_u (2\mu - 3\sigma^2 + \alpha\sigma^2 q_u) + 2k'_u)) du, \quad (18)$$

$$a_x(\tau) = \frac{(e^{-\tau\lambda} - 1) s_x}{\lambda}. \quad (19)$$

The bond price is exponential affine in the state-variable  $x_t$  and, hence, follows a stationary process. The function  $a_x(\tau)$  is the semi-elasticity of the price with respect to  $x_t$ : it is negative for any  $\tau$ , provided  $s_x > 0$ , and monotonically increasing in magnitude as a function of the residual maturity  $\tau$ .

**Proposition 4** *The equilibrium price of the equity claim on the dividend  $\delta = y^p$  with residual maturity  $\tau$  is given by:*

$$P_{t,\tau} = \delta_t e^{b_0(\tau) + b_x(\tau)x_t}, \quad 0 \leq \tau \leq T - t. \quad (20)$$

where

$$b_0(\tau) = \int_0^\tau (\beta - (\rho - 1) (\mu + (\rho - 2)\sigma^2/2) - \eta^2 b_x(u)^2/2 - b_x(u) (\bar{x}\lambda + \alpha\eta^2 h_u) - \frac{1}{2}\alpha (2\bar{x}\lambda h_u + \alpha\eta^2 h_u^2 - 2\beta k_u + q_u (2\mu + (2\rho - 3)\sigma^2 + \alpha\sigma^2 q_u) + 2k'_u)) du, \quad (21)$$

$$b_x(\tau) = \frac{(1 - e^{-\tau\lambda}) s_x (\rho - 1)}{\lambda}. \quad (22)$$

The equity claim price relative to the current dividend level is exponential affine in the state-variable  $x_t$  and, hence, follows a stationary process. The function  $b_x(\tau)$  is the semi-elasticity of the price with respect to  $x_t$ : it is positive for any  $\tau$ , provided  $s_x > 0$  and  $\rho > 1$ , and monotonically decreasing in magnitude as a function of the residual maturity  $\tau$ . Notice, that when the dividend process equals the endowment ( $\rho = 1$ ), then the equity claim price relative to the current dividend level becomes a deterministic function of the maturity (i.e.  $b_x(\tau) = 0$ ): such a result is due to the specific form of  $f$  in Eq. (5) which implies a restriction on the agent's elasticity of intertemporal substitution equal to unity.

In order to study the term-structures of interest rates and equity returns I consider both the notions of yields and the instantaneous moments of the returns. While the bond's yield is a well known quantity, defined as  $-\log(Q_{t,\tau})/\tau$ , the notion of equity yield has been introduced by [van Binsbergen, Hueskes, Koijen, and Vrugt \(2013\)](#) and is defined as  $-\log(P_{t,\tau}/\delta_t)/\tau$ .

**Corollary 1** *The yield on the discount bond with residual maturity  $\tau$  is given by:*

$$\varepsilon(t, \tau) = -\frac{1}{\tau} (a_0(\tau) + a_x(\tau)x_t). \quad (23)$$

*The instantaneous premium and volatility of the bond return are given by:*

$$(\mu_Q - r)(t, \tau) = \theta_{x,t} a_x(\tau) \eta, \quad (24)$$

$$\sigma_Q(t, \tau) = a_x(\tau) \eta. \quad (25)$$

Since the discount bond price is exponential affine in the state-variable process  $x_t$ , the bond's yield is linear in  $x_t$  and its instantaneous volatility is a deterministic function of the residual maturity  $\tau$ . Similarly also the instantaneous premium and volatility of the bond return do not depend on  $x_t$ . Such results are inherited by the homoscedasticity of  $x_t$  assumed in Eq. (7).

**Corollary 2** *The equity yield on the equity claim with residual maturity  $\tau$  is given by:*

$$p(t, \tau) = -\frac{1}{\tau} (b_0(\tau) + b_x(\tau)x_t). \quad (26)$$

*The instantaneous premium and volatility on the equity claim return are given by:*

$$(\mu_P - r)(t, \tau) = \theta_{y,t} \rho \sigma + \theta_{x,t} b_x(\tau) \eta, \quad (27)$$

$$\sigma_P(t, \tau) = \sqrt{\rho^2 \sigma^2 + b_x(\tau)^2 \eta^2}. \quad (28)$$

Similarly to the bond's yield, also the equity yield is linear in  $x_t$  and, hence, its volatility is a deterministic function of

the residual maturity  $\tau$ . Moreover, the instantaneous premium and volatility do not depend on  $y_t$  and  $x_t$ , given the homoscedasticity of  $x_t$  and  $\log y_t$  assumed in Eq. (6)-(7).

Since the instantaneous premia on both the bond and the equity returns depend on the price of risk  $\theta_{x,t}$ , then both the supply- and the demand-effects,  $s_x$  and  $d_x$ , contribute to determine the sign of the slopes of both the term-structures of premia.

The equilibrium price process of the claim on the cumulative dividend process  $\delta$  up to the final horizon  $T$  can be written as the time-integral of the prices on the equity claims:

$$P_t = \int_0^{T-t} P_{t,\tau} d\tau. \quad (29)$$

Therefore, the instantaneous premium and volatility on its return are given by

$$(\mu_P - r)(t) = \theta_{y,t} \rho \sigma + \theta_{x,t} \chi(x_t) \eta, \quad (30)$$

$$\sigma_P(t) = \sqrt{\rho^2 \sigma^2 + \chi(x_t)^2 \eta^2}, \quad (31)$$

where

$$\chi(x_t) = P_t^{-1} \partial_x P_t = P_t^{-1} \int_0^{T-t} b_x(\tau) P_{t,\tau} d\tau,$$

is the semi-elasticity of the price with respect to  $x_t$ . Both the instantaneous premium and volatility depend on the state-variable process  $x_t$ : such a time-variation is an endogenous result which obtains although the homoscedasticity of  $x_t$  and  $\log y_t$  assumed in Eq. (6)-(7). Notice that this equilibrium state-dependence of both  $(\mu_P - r)(t)$  and  $\sigma_P(t)$  is disregarded by models solved under a log-linearization of the return process, which is the case of most of long-run risk literature.

### C. The effects of demand shocks and timing preferences

By inspection of Eq. (11) for  $h_t$  we observe that it can be rewritten as

$$h_t = \frac{1}{\beta + \lambda} \left( \frac{s_x}{\lambda} (e^{-(T-t)(\beta + \lambda)} \beta + \lambda - e^{-(T-t)\beta} (\beta + \lambda)) - d_x (1 - e^{-(\beta + \lambda)(T-t)}) \right).$$

The first term on the right hand side is positive for any remaining horizon  $T - t$ , provided  $s_x > 0$ . The second term is negative if  $d_x > 0$  and vice-versa. Moreover,

$$\partial_{s_x, T-t} h_t > 0 \quad \text{and} \quad \partial_{d_x, T-t} h_t < 0,$$

that is a small changes in  $s_x$  and  $d_x$  have respectively a stronger impact respectively at long and short horizons. The term-structures of premia on bond and equity returns have slopes given by:

$$\partial_{\tau}(\mu_Q - t)(t, \tau) = \theta_{x,t} \eta \partial_{\tau} a_x(\tau),$$

$$\partial_{\tau}(\mu_P - t)(t, \tau) = \theta_{x,t} \eta \partial_{\tau} b_x(\tau),$$

such that their sign reduces to

$$\text{sign}(\partial_{\tau}(\mu_Q - t)(t, \tau)) = \text{sign}(\alpha h_t),$$

$$\text{sign}(\partial_{\tau}(\mu_P - t)(t, \tau)) = \text{sign}(-\alpha h_t),$$

provided  $s_x > 0$  and  $\rho > 1$ . Therefore, for any maturity  $\tau$ , the slopes of the two term-structures have the opposite sign.

In absence of demand shocks ( $d_x = 0$ ),  $h_t > 0$  and, hence, the sign of the slopes is characterized by the timing preferences only. If the agent has preference for the late resolution of uncertainty ( $\alpha > 0$ ), then the premia on bonds are upward sloping and those on the equity are downward sloping. The opposite relations hold if the agent has preference for the early resolution of uncertainty ( $\alpha < 0$ ). Therefore, the model cannot simultaneously generate negative slopes of the equity premia and high levels of equity premia. Recall that the premia on equity returns are

given by

$$(\mu_P - r)(t, \tau) = \sigma^2 \rho(1 - \alpha q_t) - \alpha \eta^2 h_t b_x(\tau). \quad (32)$$

For  $\alpha > 0$ , on the one hand, the slopes of equity premia are negative and, on the other hand, the first term on the right hand side of Eq. (32) is positive but lower than  $\sigma^2 \rho$ , whereas the second term is negative.

Instead, in presence of demand shocks, for  $d_x$  large enough,  $h_t$  can be negative for some horizons  $T - t$ . Hence, if the intertemporal substitution effect dominates the wealth effect ( $\alpha > 0$ ), then the premia on bonds are downward sloping and those on the equity are upward sloping. The opposite relations hold if the wealth effect dominates ( $\alpha < 0$ ). The implication for the level of the equity premia is that, for  $\alpha < 0$ , the first term on the right hand side of Eq. (32) is positive and larger than  $\sigma^2 \rho$ , whereas the second term is negative.

Under reasonable parameters, Section VI shows that the inclusion of demand shocks helps to accomodate the empirical evidence on both the level and the slopes of equity premia.

## IV. Infinite Horizon Economy

This Section considers the same full-information economy of Section III in the limiting case of infinite time-horizon. Therefore, all the results obtain by taking the limit of the results of Section III, as  $T \rightarrow \infty$ . Namely, the utility process satisfies  $\log(1 + \alpha V_t) = \alpha(q \log y_t + h x_t + k)$  for the constants

$$q = 1, \quad h = \frac{s_x - d_x}{\beta + \lambda}, \quad k = \left( \frac{\alpha(s_x - d_x)^2 \eta^2}{(\beta + \lambda)^2} + \frac{2\bar{x}(s_x - d_x)\lambda}{\beta + \lambda} + 2\mu + \sigma^2(\alpha - 1) \right) / (2\beta).$$

Hence, the utility process is monotonically increasing in  $x_t$  for  $s_x > d_x$  and vice-versa. The state-price process has dynamics  $d\pi_t = \pi_t(-r_t dt - \theta_y dB_{y,t} - \theta_x dB_{x,t})$  with

$$r_t = \beta + \mu + s_x x_t - \sigma^2(1 - \alpha),$$

and

$$\theta_y = \sigma(1 - \alpha), \quad \theta_x = \frac{\alpha \eta}{\beta + \lambda}(d_x - s_x).$$

The price for transient risk,  $\theta_y$ , is positive with  $\theta_y \leq \sigma$  if  $\alpha \geq 0$ . The price for non-transient risk,  $\theta_x$ , has the same sign of  $\alpha(d_x - s_x)$ . The prices of the discount bond and the equity claim are still

$$Q_{t,\tau} = e^{a_0(\tau) + a_x(\tau)x_t}, \quad P_{t,\tau} = \delta_t e^{b_0(\tau) + b_x(\tau)x_t}$$

where  $\delta_t = y_t^\rho$  and the constants  $q, h$  and  $k$  substitute for  $q_t, h_t$  and  $k_t$  in  $a_0(\tau), a_x(\tau), b_0(\tau)$  and  $b_x(\tau)$  from Proposition 3 and 4. Therefore, the instantaneous premia on the bond and equity claim returns are given by

$$(\mu_Q - r)(\tau) = \frac{(1 - e^{-\lambda\tau})s_x\alpha(s_x - d_x)\eta^2}{\lambda(\beta + \lambda)}, \quad (\mu_P - r)(\tau) = (1 - \alpha)\rho\sigma^2 + \frac{(1 - e^{-\lambda\tau})s_x\alpha(d_x - s_x)\eta^2(\rho - 1)}{\lambda(\beta + \lambda)}. \quad (33)$$

Such that their term-structures have slopes with sign given by

$$\text{sign}(\partial_\tau(\mu_Q - r)(\tau)) = \text{sign}(\alpha(s_x - d_x)), \quad \text{sign}(\partial_\tau(\mu_P - r)(\tau)) = \text{sign}(\alpha(d_x - s_x)), \quad (34)$$

provided  $\rho > 1$ . The infinite-horizon economy clearly shows how the shape of the equilibrium term-structures depends on the interaction of the timing preferences ( $\alpha \geq 0$ ) and demand shocks ( $d_x \geq s_x$ ), since the sign of  $h$  is given by the sign of  $s_x - d_x$ , whereas that of  $h_t$  in Proposition 1 also depends on the residual time-horizon  $T - t$ , as discussed in Section III.

## V. Incomplete Information Economy

This section investigates the case of incomplete information, in which the agent does not observe the state-variable process  $x$ . While most of long-run risk literature disregards the latent nature of  $x$ , I characterize conditions under which incomplete information is important to model the equilibrium term-structures. Namely, I am interested into show how the quality of available information, the timing preferences and demand shocks jointly affect the equilibrium term-structures. All results are derived in the Appendix. For the sake of exposition I consider the case of the infinite-horizon economy ( $T = \infty$ ), but all results can be easily generalized to the finite-horizon case.

## A. Information structure and learning

Let  $\mathbf{F}^n = \{\mathcal{F}_t^n : t \geq 0\}$  be a filtration stochastically independent of the Brownian motion  $\mathbf{B} = (B_y, B_x)'$ : namely I introduce a Brownian motion  $Z$  which is independent of  $\mathbf{B}$  and I let  $\mathbf{F}^n$  be the filtration generated by  $Z$ . Denote with  $\mathbf{F}^o = \{\mathcal{F}_t^o : t \geq 0\}$  the filtration observed by the agent: as long as  $\mathcal{F}_t^o = \mathcal{F}_t^{y,x} \vee \mathcal{F}_t^n$ , the additional information  $\mathcal{F}_t^n$  is irrelevant to consumption and, hence,  $\mathbf{F}^n$  does not affect prices in equilibrium. Instead, the “noisy” filtration  $\mathbf{F}^n$  can play a role when the agent does not observe  $x$ .

Consider an economy that is identical to that described in the previous sections, with the exception that the agent does not observe  $x$ . Instead, the agent observes a signal process  $s$  with dynamics:

$$ds_t = x_t dt + \mathbf{v}_s' d\bar{\mathbf{B}}_t \quad (35)$$

where  $\bar{\mathbf{B}} = (B_y, B_x, Z)'$  is a three-dimensional Brownian motion and  $\mathbf{v}_s = (\mathbf{v}_{sy}, \mathbf{v}_{sx}, \mathbf{v}_{ss})' \in \mathbb{R}^3$ . The total information observed by the agent is given by the filtration  $\mathcal{F}_t^o = \mathcal{F}_t^{y,s} = \{\mathcal{F}_t^{y,s} : t \geq 0\}$ , generated by the endowment process  $y$  and the signal  $s$ . Therefore, the agent of this economy receives less timely information, which is “contaminated” by  $Z$  for  $\mathbf{v}_{ss} \neq 0$ . In particular, since the covariances of  $s$  with  $y$  and  $x$  do not depend on  $\mathbf{v}_{ss}$ , such a parameter can be interpreted as a measure of the quality of information available to the agent.

Consider now the agent’s learning problem. Let introduce the following notation:  $\mathbf{v}_y = (\sigma, 0, 0)'$  and  $\mathbf{v}_x = (0, \eta, 0)'$  and define  $\mathbf{v} = (\mathbf{v}_y, \mathbf{v}_x, \mathbf{v}_s)$ . I assume that  $\mathbf{v}'\mathbf{v}$  is positive definite and, hence, the matrix  $\Sigma = (\mathbf{v}_y, \mathbf{v}_s)'(\mathbf{v}_y, \mathbf{v}_s)$  is positive definite too. Let the conditional mean and variance of the state-variable process  $x$  on the agent’s available information given by respectively:

$$\hat{x}_t = \mathbb{E}[x_t | \mathcal{F}_t^o] = \mathbb{E}[x_t | \mathcal{F}_t^{y,s}], \quad t \geq 0,$$

and

$$\hat{\phi}_t = \mathbb{E}[(x_t - \hat{x}_t)^2 | \mathcal{F}_t^o] = \mathbb{E}[(x_t - \hat{x}_t)^2 | \mathcal{F}_t^{y,s}], \quad t \geq 0.$$

Standard filtering theory provides the result that the distribution of  $x_t$  conditional on the agent’s information  $\mathcal{F}_t^o$  is Gaussian.

**Proposition 5** *The following results hold.*

i) *The conditional variance  $\hat{\phi}$  is a deterministic function of time with steady-state limit given by:*

$$\hat{\phi}^\infty = \frac{\eta^2 \sigma v_{ss}^2}{\lambda \sigma (v_{ss}^2 + v_{sx}^2) + \eta v_{sx} (\sigma - s_x v_{sy}) + \sqrt{v_{ss}^2 + v_{sx}^2} \sqrt{(\lambda^2 \sigma^2 + \eta^2 s_x^2) v_{ss}^2 + (\sigma(\eta + \lambda v_{sx}) - \eta s_x v_{sy})^2}}.$$

ii) *For any  $t \geq 0$ , the following dynamics hold*

$$d\hat{x}_t = \lambda(\bar{x} - \hat{x}_t)dt + (s_x \hat{\phi}_t, \hat{\phi}_t + \eta v_{sx}) \Sigma^{-1/2} d\mathbf{W}_t,$$

with  $\hat{x}_0 = x_0$  and

$$\begin{pmatrix} dy_t/y_t \\ ds_t \end{pmatrix} = \begin{pmatrix} \mu + s_x \hat{x}_t \\ \hat{x}_t \end{pmatrix} dt + \Sigma^{1/2} d\mathbf{W}_t,$$

where  $\mathbf{W} = (W_y, W_s)'$  is a standard Brownian motion under  $\mathbf{F}^o = \mathbf{F}^{y,s}$ .

iii) *The filtration generated by  $\mathbf{W}$  is  $\mathbf{F}^{y,s}$ .*

Proposition 5 reduces the agent's problem to one of complete information, given the dynamics under  $\mathbf{F}^o$ , as in [Dothan and Feldman \(1986\)](#) among others. Therefore, the standard asset pricing methodology of the previous sections naturally adapts to the current setting of incomplete information.

The focus of the analysis concerns the role of information quality, as represented by the parameter  $v_{ss}$ . Hence, for the only purpose of exposition, I make the following simplifying assumptions. I set  $v_{ss} > 0$ ,  $v_{sy} = v_{sx} = 0$  and I let the conditional variance of  $x$  be equal to its steady-state limit  $\hat{\phi}^\infty$ , as in [Dumas, Kurshev, and Uppal \(2009\)](#). That is, I assume a long enough horizon of past observations, so that the posterior variance has converged towards its steady-state. As a consequence the dynamics of  $\hat{x}$ ,  $y$  and  $s$  satisfy:

$$d\hat{x}_t = \lambda(\bar{x} - \hat{x}_t)dt + \sigma_{xy} dW_{y,t} + \sigma_{xs} dW_{s,t}, \quad (36)$$

$$dy_t/y_t = (\mu + s_x \hat{x}_t)dt + \sigma dW_{y,t}, \quad (37)$$

$$ds_t = \hat{x}_t dt + v_{ss} dW_{s,t}, \quad (38)$$

where

$$\sigma_{xy} = \frac{\eta^2 s_x v_{ss}}{\lambda \sigma v_{ss} + \sqrt{\eta^2 \sigma^2 + (\lambda^2 \sigma^2 + \eta^2 s_x^2) v_{ss}^2}}, \quad \sigma_{xs} = \frac{\eta^2 \sigma}{\lambda \sigma v_{ss} + \sqrt{\eta^2 \sigma^2 + (\lambda^2 \sigma^2 + \eta^2 s_x^2) v_{ss}^2}}. \quad (39)$$

Notice that  $\sigma_{xy} = \hat{\phi}^\infty s_x / \sigma$  and  $\sigma_{xs} = \hat{\phi}^\infty / v_{ss}$  are respectively increasing and decreasing in  $v_{ss}$ . Moreover, for  $v_{ss} \rightarrow \infty$ ,  $\sigma_{xy} \rightarrow \frac{s_x \eta^2}{\lambda \sigma + \sqrt{s_x^2 \eta^2 + \lambda^2 \sigma^2}}$  and  $\sigma_{xs} \rightarrow 0$ , whereas for  $v_{ss} \rightarrow 0$ ,  $\sigma_{xy} \rightarrow 0$  and  $\sigma_{xs} \rightarrow \eta$ . In the former case the signal  $s$  is too noisy to be useful, whereas in the latter case the signal  $s$  is precise enough to recover the dynamics for  $x$  of the full information economy.

## B. Value function, state-price density and equilibrium security prices

Given the dynamics in Eq. (36)-(37) and  $T = \infty$ , the explicit utility formula is similar to that of Section IV. Namely, the utility process satisfies  $\log(1 + \alpha V_t) = \alpha(q \log y_t + h \hat{x}_t + k)$  for the constants

$$q = 1, \quad h = \frac{s_x - d_x}{\beta + \lambda}, \quad k = \frac{\mu}{\beta} + \frac{(\alpha - 1)\sigma^2}{2\beta} + \frac{(s_x - d_x)((\sigma_{xs}^2 + \sigma_{xy}^2)\alpha(s_x - d_x) + 2\bar{x}\lambda(\beta + \lambda) + 2\sigma_{xy}\alpha(\beta + \lambda)\sigma)}{2\beta(\beta + \lambda)^2}.$$

Hence, the utility process is monotonically increasing in  $x_t$  for  $s_x > d_x$  and vice-versa. The state-price process has dynamics

$$\frac{d\pi_t}{\pi_t} = -r_t dt - \theta_y dW_{y,t} - \theta_s dW_{s,t}.$$

The instantaneous interest rate is given by

$$r_t = \beta + \mu + s_x \hat{x}_t - \sigma^2(1 - \alpha) + \frac{\alpha \sigma_{xy}}{\beta + \lambda}(s_x - d_x),$$

where the last term is due to incomplete information and has the same sign of  $\alpha(s_x - d_x)$ . The equilibrium prices of risk are

$$\theta_y = \sigma(1 - \alpha) + \frac{\alpha \sigma_{xy}}{\beta + \lambda}(d_x - s_x), \quad \theta_s = \frac{\alpha \sigma_{xs}}{\beta + \lambda}(d_x - s_x).$$

Notice that now  $\theta_y$  is given by two components:  $\sigma(1 - \alpha)$  is the usual price for transient risk, whereas  $\theta_x - \sigma(1 - \alpha)$  is a price for non-transient risk with sign equal to that of  $\alpha(d_x - s_x)$ . Such a term obtains under incomplete information because the agent's filtered dynamics of the state-variable process  $\hat{x}$  depends on both  $W_{y,t}$  and  $W_{s,t}$ . The price of risk,

$\theta_s$ , is an additional price for non-transient risk and has the same sign of  $\alpha(d_x - s_x)$ . The effect of the parameter  $d_x$  on the instantaneous interest rate has the opposite sign than its effect on the prices of risk. Therefore, demand shocks help to explain the low risk-free rate and the high Sharpe ratio of the stock markets, provided  $\alpha < 0$  and  $d_x > s_x$ .

The prices of the discount bond and the equity claim are still

$$Q_{t,\tau} = e^{a_0(\tau) + a_x(\tau)\hat{x}_t}, \quad P_{t,\tau} = \delta_t e^{b_0(\tau) + b_x(\tau)\hat{x}_t}$$

where  $\delta_t = y_t^p$  and the deterministic functions  $a_0(\tau)$ ,  $a_x(\tau)$ ,  $b_0(\tau)$  and  $b_x(\tau)$  are derived in the Appendix. Therefore, the instantaneous premia on the bond and equity claim returns are given by

$$(\mu_Q - r)(\tau) = \frac{\sigma(1-\alpha)(e^{-\lambda\tau}-1)s_x\sigma_{xy}}{\lambda} + \frac{(1-e^{-\lambda\tau})s_x\alpha(s_x-d_x)(\sigma_{xy}^2 + \sigma_{xs}^2)}{\lambda(\beta+\lambda)}, \quad (40)$$

$$(\mu_P - r)(\tau) = (1-\alpha)\rho\sigma^2 + \frac{\sigma(1-\alpha)(1-e^{-\lambda\tau})s_x\sigma_{xy}(\rho-1)}{\lambda} + \frac{(1-e^{-\lambda\tau})s_x\alpha(d_x-s_x)(\sigma_{xy}^2 + \sigma_{xs}^2)(\rho-1)}{\lambda(\beta+\lambda)}. \quad (41)$$

The first term on the right and side of Eq. (40) and the second term on the right hand side of Eq. (41) are due to incomplete information. Namely, these terms represent a remuneration for the instantaneous covariance among the consumption process and the state-variable process, as estimated by the agent. Notice that such remunerations of the discount bond and the equity claim are respectively negative and positive (provided  $\rho > 1$ ), despite the timing preferences (i.e. for any  $\alpha < 1$ ).

### C. The effects of information quality, demand shocks and timing preferences

In order to study the role of information quality and its interaction with the demand shocks and the timing preferences, I focus on the parameter  $v_{ss}$  which represents the instantaneous volatility of the signal,  $s$ . Therefore, the lower the parameter  $v_{ss}$ , the more precise the signal and the larger the quality of available information or, equivalently, the more timely the arrival of useful information. Instead, the larger the parameter  $v_{ss}$ , the more “noisy” the economy. Let consider at first the utility process  $V$ .

**Corollary 3** *The derivative of the utility process,  $V$ , with respect to the instantaneous volatility of the signal,  $v_{ss}$ , is*

given by

$$\frac{\partial V_t}{\partial v_{ss}} = \frac{\alpha(s_x - d_x)(s_x \beta + d_x \lambda) \sigma^3 \eta^4 e^{\alpha(q \log y_t + h \hat{x}_t + k)}}{\beta(\beta + \lambda)^2 \sqrt{s_x^2 \eta^2 v^2 + (\eta^2 + \lambda^2 v^2) \sigma^2} \left( \lambda v \sigma + \sqrt{s_x^2 \eta^2 v^2 + (\eta^2 + \lambda^2 v^2) \sigma^2} \right)^2}. \quad (42)$$

Therefore, we have that  $\text{sign}(\partial_{v_{ss}} V_t) = \text{sign}(\alpha(s_x - d_x))$ . In absence of demand shocks ( $d_x = 0$ ), the utility process is increasing (decreasing) in information quality for  $\alpha < 0$  ( $\alpha > 0$ ). That is, when the agent has preference for the early resolution of uncertainty his utility is increasing in the information quality. Vice-versa when the agent has preference for the late resolution of uncertainty, its utility is decreasing in the information quality. In absence of demand shocks, such a relationship between the effects of the observed filtration and the quality of information is fully characterized by the timing preferences ( $\alpha \geq 0$ ).

The rationale is the following: an increase in  $v_{ss}$  leads to an increase in  $\sigma_{xy}$  and a decrease in  $\sigma_{xs}$ . Hence, the filtered state-process becomes more instantaneously correlated with the endowment process. As a consequence, the additional uncertainty due to an increase in  $v_{ss}$  is more penalized under preference for the early resolution of uncertainty. Indeed, for  $\alpha < 0$ , the prices of risk  $\theta_y$  and  $\theta_s$  respectively increase and decrease. As a consequence, an increase in  $v_{ss}$  respectively increases and decreases the exposition of  $\hat{x}$  to the shocks  $W_y$  and  $W_s$  when the associated prices of risk increase and decrease as well. The opposite holds for  $\alpha > 0$ .

Instead, when demand shocks are large enough ( $d_x > s_x$ ), the utility process becomes increasing (decreasing) in information quality when the agent has preference for the late (early) resolution of uncertainty. Indeed, even if the aggregator function is convex in the utility argument for  $\alpha < 0$ ,<sup>7</sup> the perceived consumption  $\tilde{c}$ , as perturbed by the demand shocks, can alter the effect of the information quality. Namely, a strong demand-effect ( $d_x > s_x$ ) highly penalizes the states of the world in which expected growth is high (recall that  $\mathbb{E}[dy_t | \mathcal{F}_t^o]/y_t = (\mu + s_x \hat{x}_t)dt$ ). As a consequence, the prices of non-transient risk,  $\theta_y - \sigma(1 - \alpha)$  and  $\theta_s$ , are respectively decreasing and increasing in  $v_{ss}$  for  $\alpha < 0$ . Therefore, even if an increase in  $v_{ss}$  leads to an increase of the instantaneous correlation of  $\hat{x}$  with the endowment process, the expositions of  $\hat{x}$  to the shocks  $W_y$  and  $W_s$  move in the opposite direction than the corresponding prices of risk.

Let consider now the term-structures of premia on bond and equity returns. As discussed in Section IV, Eq. (34) shows that under full information the slopes of the term-structures of premia have sign fully characterized by the

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<sup>7</sup>Recall that  $f_{vv}(\tilde{c}, v) = -\alpha\beta/(1 + \alpha v)$ , with  $1 + \alpha v > 0$ .

interaction of the timing preferences ( $\alpha \geq 0$ ) with the demand shocks ( $d_x \geq s_x$ ). Instead, under incomplete information premia have more involved slopes. The first term on the right hand side of Eq. (40) and the second term on the right hand side of Eq. (41) depend on the maturity but are respectively negative and positive, despite both timing preferences and demand-shocks.

Namely, the slopes of term-structures of premia on the discount bond and the equity claim returns are given by

$$\partial_\tau(\mu_Q - r)(\tau) = e^{-\lambda\tau} s_x \left( \alpha(s_x - d_x) \frac{\sigma_{xy}^2 + \sigma_{xs}^2}{\beta + \lambda} - (1 - \alpha) \sigma \sigma_{xy} \right), \quad (43)$$

$$\partial_\tau(\mu_P - r)(\tau) = e^{-\lambda\tau} s_x \left( \alpha(d_x - s_x) \frac{\sigma_{xy}^2 + \sigma_{xs}^2}{\beta + \lambda} + (1 - \alpha) \sigma \sigma_{xy} \right) (\rho - 1). \quad (44)$$

The second term in the parenthesis for both bond and equity is due to incomplete information and is respectively negative and positive. For  $\alpha < 0$ , positive and negative slopes respectively for bond and equity premia obtain only if the demand-effect ( $d_x - s_x$ ) is large enough in magnitude to more than offset the terms due to incomplete information.

Now I focus on the role of information quality. The instantaneous volatility of the signal  $v_{ss}$  enters both the level and the slope of the term-structures of premia through the posterior variances  $\sigma_{xy}^2$  and  $\sigma_{xs}^2$ . In particular, it is interesting to verify whether a change in information quality can lead to a change in the sign of the slopes of bond and equity premia. Moreover, the conditions on both timing preferences and demand shocks under which such a change in the slopes can obtain tell us when the arrival of information is crucial to the equilibrium determination of the term-structures.

**Corollary 4** *The following results hold:*

i) for  $\alpha(d_x - s_x) \geq 0$ , the sign of the slope of bond premia does not depend on  $v_{ss}$ ,

ii) for  $\alpha(d_x - s_x) < 0$ ,

$$\partial_\tau(\mu_Q - r)(\tau) \begin{cases} > 0 & v_{ss} < v_{ss}^*, \\ = 0 & v_{ss} = v_{ss}^*, \\ < 0 & v_{ss} > v_{ss}^*, \end{cases}$$

iii) for  $\alpha(d_x - s_x) \geq 0$ , the sign of the slope of equity premia does not depend on  $v_{ss}$ ,

iv) for  $\alpha(d_x - s_x) < 0$ ,

$$\partial_{\tau}(\mu_P - r)(\tau) \begin{cases} > 0 & v_{ss} > v_{ss}^*, \\ = 0 & v_{ss} = v_{ss}^*, \\ < 0 & v_{ss} < v_{ss}^*, \end{cases}$$

where the threshold  $v_{ss}^*$  is given by

$$v_{ss}^* = \frac{(s_x - d_x)\alpha\eta\sigma}{\sqrt{-\alpha^2\eta^2(d_x - s_x)^2s_x^2 + (\alpha - 1)(\beta + \lambda)s_x\sigma^2(2\alpha\lambda d_x - (\beta - \alpha\beta + \lambda + \alpha\lambda)s_x)}},$$

and for  $\alpha(d_x - s_x) < 0$ ,

$$\partial_{\gamma}v_{ss}^* = \begin{cases} > 0 & \alpha > 0, \\ < 0 & \alpha < 0, \end{cases}$$

with  $\gamma = 1 - \alpha$  denoting the agent's relative risk aversion.

Corollary 4 provides a number of interesting insights. Information quality can alter the sign of slopes of bond and equity premia only for  $\alpha(d_x - s_x) < 0$ , that is when the term-structures of premia on bond and equity returns are respectively increasing and decreasing –which is the case observed in the real data. Moreover, the condition reduces to  $\alpha > 0$  in absence of demand shocks ( $d_x = 0$ ). However, this is a case of low empirical interest, since for  $\alpha > 0$  the agent's relative risk aversion is lower than one and the model cannot generate a sizeable equity premium under reasonable parameters. Therefore, in absence of demand shocks the role of information quality is quite limited.

Instead, when the demand-effect is large enough ( $d_x > s_x$ ), the condition  $\alpha(d_x - s_x) < 0$  is consistent with both relative risk aversion larger than unity ( $\alpha < 0$ ) and downward sloping term-structures of equity premia under full information (see Eq. (34)). If information is incomplete, there exists a threshold  $v_{ss}^*$  such that: for high information quality ( $v_{ss} < v_{ss}^*$ ) the signs of the slopes from the full information economy are preserved, whereas for low information quality ( $v_{ss} > v_{ss}^*$ ) the signs are reverted. For  $v_{ss} = v_{ss}^*$ , the term-structures are flat. Therefore, for  $\alpha(d_x - s_x) < 0$  the equilibrium term-structures have the right shape under full information, but such a shape is preserved under incomplete information only if a minimum requirement ( $v_{ss} < v_{ss}^*$ ) on information quality is satisfied.

Finally, if the agent has preference for the early (late) resolution of uncertainty, an increase in risk aversion leads to a decrease (increase) in the threshold  $v_{ss}^*$ . Hence, an increase in risk aversion also leads to an increase (decrease) in the quality of available information needed to preserve the sign of the slopes from the full information economy.

## VI. Numerical Illustration and Discussion

In order to provide a numerical illustration of the quantitative results of the model, I consider the infinite horizon economies of Section IV and V respectively under full and incomplete information. Although the model is voluntarily kept simple and parsimonious, some of the quantitative implications are of empirical interest.

In order to analyse the joint role of the timing preferences, the demand shocks and the information quality, the analysis focuses on the parameters  $\alpha$ ,  $d_x$  and  $v_{ss}$ . For the purpose of comparison, I consider as a benchmark the model without demand shocks ( $d_x = 0$ ), which is very similar to a long-run risk model under the restriction imposed by the aggregator function  $f(c, v)$  in Eq. (5), which implies an elasticity of intertemporal substitution equal to one.<sup>8</sup>

The model is calibrated as follows:  $\bar{x}$  and  $s_x$  are standardized respectively to zero and one for the sake of simplicity; then I set  $\mu = 1.5\%$  and  $\sigma = 3.5\%$  to capture the first two moments of aggregate consumption. The state-variable process is assumed to be smooth and quite persistent: hence, in line with most of long-run risk literature, I set  $\lambda = .25$  and  $\eta = 1\%$ . The leverage parameter is set to  $\rho = 4.5$  to match the excess volatility of aggregate dividends over aggregate consumption.

Given the above parameters, I investigate the model's implications for  $\alpha \in (-9, 1)$ , which implies a relative risk aversion  $\gamma \in (0, 10)$  in the usual range of values considered in the literature. Conditional on the level of  $\alpha$ , the demand-effect parameter  $d_x$  can be set to balance both the level and the slope of equity premia implied by the model. The time-discount rate  $\beta$  can be set, as usual, to match the unconditional level of the risk-free rate given all the other parameters. Finally, I let  $v_{ss}$  to vary in an interval of  $v_{ss}^*$  in order to study the effect of information quality on the slopes of the term-structures. The baseline model calibration is summarized in Table I.

Insert Table I about here.

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<sup>8</sup>Notice that such an assumption is quite conservative: indeed, there is no consensus in the literature whether the elasticity of intertemporal substitution should be above or below unity.

Consider at first the full information economy. Table II shows the main asset pricing moments under several scenarios. Namely, I study the six combinations which obtain from  $\alpha \geq 0$  and  $d_x \leq s_x$ .

Insert Table II about here.

The unconditional level of the risk-free rate does not depend on the demand-effect parameter  $d_x$  but slightly varies with the timing preferences: in particular, it is increasing with  $\alpha$  and reaches the empirical level, somewhat below 1%, for  $\alpha = -9$ , which implies the standard case in the literature of relative risk aversion  $\gamma = 10$ . The instantaneous volatility of the risk-free rate is given by  $s_x \eta$  and, hence, is independent of preference parameters. In the calibration this leads to a low volatility of 1%, which captures the link between the smooth dynamics of the interest rate and of the macroeconomic fundamentals, as in the real data.

Consider now the equity premium of Eq. (30), for  $T \rightarrow \infty$ . The model does not generate a sizeable market compensation under both preference for the late resolution of uncertainty ( $\alpha > 0$ ) and no preference for the resolution of uncertainty ( $\alpha = 0$ ). In these cases, the premium is one order of magnitude smaller than its empirical counterpart (about 0.5% versus 6%), despite the strength of demand shocks. Instead, a sizeable premium can obtain under preference for the early resolution of uncertainty. In particular, in absence of demand shocks, a relative risk aversion equal to 10 (that is,  $\alpha = -9$ ) leads to an equilibrium compensation (about 10%) substantially larger than in the real data. This is due to the fact that both the terms in Eq. (30) are positive. In presence of demand shocks ( $s_x < d_x = 1.1$ ), the equity premium is smaller but quite close to its empirical counterpart for a relative risk aversion of 10 (about 5%). The level of the stock return volatility of Eq. (31) is slightly sensitive to the preference parameters and is close to the observed volatility (about 21% versus about 18-20%). As a consequence, a high Sharpe ratio only obtains under preference for the early resolution of uncertainty. The latter is somewhat lower than in real data for  $\alpha = -9$  and  $d_x = 1.1$  (about 24% versus 30-35%), but this can be imputed to the simplicity of the model which, for instance, does not account for the heteroscedasticity of fundamentals and their pricing at equilibrium.

When the agent has no preference for the timing resolution of uncertainty ( $\alpha = 0$ ), the premia on the discount bonds are equal to zero, despite demand shocks. Instead, in absence of demand shocks ( $d_x = 0$ ), long-term bonds deserve a premium which is positive and negative respectively under preference for the late and the early resolution of uncertainty. Hence, for  $d_x = 0$ , the positive term premium, which is observable in the real data, can obtain only for

$\alpha > 0$ , that is when the model fails to command a high equity premium. Vice-versa, when the demand-effect is large enough ( $s_x < d_x = 1.1$ ), bonds deserve a premium which is positive and negative respectively under preference for the early and the late resolution of uncertainty. Therefore, a positive term premium obtains when the model generates a sizeable equity premium.

The term-structures of premia on both bond and equity returns are flat when the agent has no preference for the timing resolution of uncertainty ( $\alpha = 0$ ). In absence of demand shocks ( $d_x = 0$ ), the slopes for bond and equity premia are respectively positive and negative under preference for the late resolution of uncertainty. The opposite signs obtain under preference for the early resolution of uncertainty. Hence, for  $d_x = 0$ , the model cannot reconcile a high equity premium with the correct specification of the term-structures. Instead, when the demand-effect is large enough ( $s_x < d_x = 1.1$ ) and the agent has preference for the early resolution of uncertainty ( $\alpha < 0$ ), the model simultaneously leads to a high equity premium and the right shape of the term-structures. Indeed, positive and negative slopes for respectively bond and equity premia obtain, in line with the empirical evidence and the recent works by [van Binsbergen et al. \(2012\)](#) and [van Binsbergen et al. \(2013\)](#).

Figure 1 shows the equity premium as a function of timing preferences ( $\alpha$ ), in the cases of absence of demand shocks ( $d_x = 0$ , blue line) and of large demand shocks ( $s_x < d_x = 1.1$ , red line). The shaded areas indicate where the premia are compatible with positive and negative slopes for the term-structures of premia on bond and equity returns.

Insert Figure 1 about here.

Figure 2 shows the premia on bond (left panels) and equity (right panels) returns as a function of the maturity ( $\tau$ ) and the timing preferences ( $\alpha$ ), in the cases of absence of demand shocks ( $d_x = 0$ , upper panels) and of large demand shocks ( $s_x < d_x = 1.1$ , lower panels).

Insert Figure 2 about here.

It clearly appears that demand shocks are necessary to reconcile the level and the slopes of equity premia.

Consider now the incomplete information economy. Table III shows the main asset pricing moments under several scenarios, similarly to Table II.

Insert Table III about here.

The panel A of Table III considers the case of high information quality: the volatility of the noisy signal is set to a value ( $v_{ss} = .1\%$ ) lower than the threshold  $v_{ss}^*$ .<sup>9</sup> As a consequence, the model preserves the correct specification of the term-structures of the full information economy in the two scenarios in which  $\alpha(d_x - s_x) < 0$ . That is, when either the agent has preference for the late resolution of uncertainty ( $\alpha > 0$ ) and the demand effect is little ( $d_x < s_x$ ) or the agent has preference for the early resolution of uncertainty ( $\alpha < 0$ ) and the demand effect is large ( $d_x > s_x$ ). Recall that the model leads to a sizeable equity premium only in the latter case and under both full and incomplete information. The panel B of Table III considers the case of low information quality: the volatility of the noisy signal is set to a value ( $v_{ss} = 5\%$ ) higher than the threshold  $v_{ss}^*$ . As a consequence, even for  $\alpha(d_x - s_x) < 0$ , the model fails to produce the correct specification of the term-structures.

Figure 3 shows the premia on bond (left panels) and equity (right panels) returns as a function of the maturity ( $\tau$ ) and the information quality ( $v_{ss}$ ), in the cases of absence of demand shocks ( $d_x = 0$ , upper panels) and of large demand shocks ( $s_x < d_x = 1.1$ , lower panels), for  $\alpha = -9$ .

Insert Figure 3 about here.

Since  $\alpha < 0$ , in presence of demand shocks we can observe the change in the sign of the slopes when  $v_{ss}$  increases above the threshold  $v_{ss}^* = .3\%$ . Instead, in absence of demand shocks we can observe that the term-structures are monotone with information quality.

Two model's implications are noteworthy. First, standard asset pricing facts –such as the first two return moments of both the risk-free asset and the stock market– are little sensitive to the degree of information quality. Indeed, for given  $\alpha$  and  $d_x$ , the steady-state moments in Table II and in both the panels of Table III are very similar. Second, in the only case in which the model reconciles the standard asset pricing facts with the correct specification of the term-structures ( $\alpha < 0$  and  $d_x > s_x$ ), the shapes of the term-structures are highly sensitive to the degree of information quality. Therefore, if we only look at the standard asset pricing facts, then incomplete information is not particularly important and can maybe be disregarded at evaluating an equilibrium asset pricing model. Instead, modelling the information structure and quality (or equivalently the timely arrival of useful information) becomes crucial when also a term-structure perspective is taken into account.

<sup>9</sup>The threshold  $v_{ss}^*$  is relevant under the condition  $\alpha(d_x - s_x) < 0$ , which is satisfied in Table III for  $\alpha = .5$  and  $d_x = 0$  and for  $\alpha = -9$  and  $d_x = 1.1$ . The corresponding thresholds are respectively given by  $v_{ss}^* = 3\%$  and  $v_{ss}^* = .3\%$ .

## VII. Conclusion

Leading equilibrium models cannot simultaneously explain premia and volatility of bond and equity returns as well as their term-structures of premia. This paper argues that demand shocks play a crucial role in asset pricing and that can help to reconcile the above stylized facts. Namely, demand shocks are introduced as the source of valuation risk, associated to future cash-flows, due to the perception of utility from consumption, conditional to the state of the economy. Under stochastic differential utility, demand shocks can alter the sign of the equilibrium prices of (non-transient) risk and, hence, can reverse the sign of the slopes of the equilibrium term-structure, for given timing preferences.

The interaction of demand shocks and timing preferences is analytically characterized under both full and incomplete information. In the latter case, the minimum requirement on the degree of information quality, which preserves the correct specification of the term-structures, is derived. While information quality barely affects the unconditional moments of equity returns, it is crucial to their term-structure properties.

The closed-form framework can be easily generalized to more general settings of exogenous uncertainty and different specifications of demand shocks. Here, the model is kept simple and parsimonious for the sake of exposition and direct comparison with the seminal equilibrium model by [Bansal and Yaron \(2004\)](#). A calibration shows that the model captures many asset pricing moments under reasonable parameters and supports the model's assumption about demand shocks.

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## Appendix

### Finite-horizon, full information economy

**Proposition A** *Under the dynamics in Eq. (6)-(7)-(8), we have that for any  $0 \leq t \leq T$ ,*

$$\mathbb{E} [\tilde{c}_{t+\tau}^a \mid \mathcal{F}_t^{y,\mathbf{x}}] = y_t^a e^{m_0(\tau) + m_x(\tau)x_t}, \quad 0 \leq \tau \leq T - t,$$

with

$$\begin{aligned} m_0(\tau) &= \frac{a}{4\beta^2\lambda^3} \left( e^{-2\lambda\tau} \left( -a\eta^2 (s_x\beta + d_x\lambda)^2 + 4e^{\lambda\tau}\beta(s_x\beta + d_x\lambda) (as_x\eta^2 + \bar{x}\lambda^2) \right) \right. \\ &\quad \left. + (-(s_x\beta + d_x\lambda) (4\bar{x}\beta\lambda^2 + a\eta^2(3s_x\beta - d_x\lambda)) + 2\beta^2\lambda (as_x^2\eta^2 + 2s_x\bar{x}\lambda^2 + \lambda^2(2\mu + (a-1)\sigma^2))\tau) \right), \\ m_x(\tau) &= a \left( (1 - e^{-\lambda\tau}) \frac{s_x}{\lambda} - e^{-\lambda\tau} \frac{d_x}{\beta} \right), \end{aligned}$$

for any  $a \geq 0$  such that the expectation exists finite. Moreover,

$$\frac{\partial_x \mathbb{E} [\tilde{c}_{t+\tau}^a \mid \mathcal{F}_t^{y,\mathbf{x}}]}{\mathbb{E} [\tilde{c}_{t+\tau}^a \mid \mathcal{F}_t^{y,\mathbf{x}}]} - \frac{\partial_x \mathbb{E} [c_{t+\tau}^a \mid \mathcal{F}_t^{y,\mathbf{x}}]}{\mathbb{E} [c_{t+\tau}^a \mid \mathcal{F}_t^{y,\mathbf{x}}]} \leq 0, \quad \forall d_x \geq 0.$$

**Proof of Proposition A:** Let consider the conditional expectation:

$$\mathbb{E} [\tilde{c}_{t+\tau}^a \mid \mathcal{F}_t^{y,\mathbf{x}}], \quad 0 \leq \tau \leq T - t,$$

and guess a trial solution:  $M(t, y_t, x_t) = y_t^a e^{m_0(\tau) + m_x(\tau)x_t}$ . Then, an application of Itô's lemma to  $M$  gives the following partial differential equation

$$\begin{aligned} 0 &= M_t + M_y \mathfrak{D}y + \frac{1}{2} M_{y,y} \langle y \rangle + M_x \mathfrak{D}x + \frac{1}{2} M_{x,x} \langle x \rangle \\ &= m'_0(\tau) + m'_x(\tau)x + a(\mu + s_x x) + a(a-1)\sigma^2/2 + m_x(\tau)\lambda(\bar{x} - x) + (m_x(\tau)\eta)^2/2, \end{aligned}$$

with  $m_0(0) = 0$  and  $m_x(0) = -a(d_x/\beta)$  and where  $\mathfrak{D}$  is the infinitesimal drift operator. The candidate solution solves the partial differential equation if and only if

$$\begin{aligned} 0 &= m'_x(\tau) + as_x - m_x(\tau)\lambda, \\ 0 &= m'_0(\tau) + a\mu + a(a-1)\sigma^2/2 + m_x(\tau)\lambda\bar{x} + (m_x(\tau)\eta)^2/2, \\ 0 &= m_0(0) = m_x(0) + a(d_x/\beta). \end{aligned}$$

Hence,  $\mathbb{E}[\tilde{c}_{t+\tau}^a | \mathcal{F}_t^{y,x}] = M(t, y_t, x_t)$ , once we solve for  $m_0(\tau)$  and  $m_x(\tau)$ . By setting  $d_x = 0$ , we have  $\mathbb{E}[c_{t+\tau}^a | \mathcal{F}_t^{y,x}] = M(t, y_t, x_t)$ .

It easily follows that

$$\frac{\partial_x \mathbb{E}[\tilde{c}_{t+\tau}^a | \mathcal{F}_t^{y,x}]}{\mathbb{E}[\tilde{c}_{t+\tau}^a | \mathcal{F}_t^{y,x}]} - \frac{\partial_x \mathbb{E}[c_{t+\tau}^a | \mathcal{F}_t^{y,x}]}{\mathbb{E}[c_{t+\tau}^a | \mathcal{F}_t^{y,x}]} = -ae^{-\tau}(d_x/\beta) \leq 0,$$

for any  $d_x \geq 0$ . *q.e.d.*

**Proof of Proposition 1:** An application of Itô's lemma to the expression for  $V$  gives the following partial differential equation for  $V_t = J(t, y_t, x_t)$ :

$$0 = J_t + J_y(\mu + s_x x)y + \frac{1}{2}J_{y,y}\sigma^2 y^2 + J_x\lambda(\bar{x} - x) + \frac{1}{2}J_{x,x}\eta^2 + f(\tilde{c}, J)$$

with  $\tilde{c} = \mathfrak{C}(y, x)$ . A trial solution for  $J$  is

$$\log(1 + \alpha J)/\alpha = (q_t \log y + h_t x + k_t).$$

Notice that as long as  $\alpha \rightarrow 0$ , the left hand side converges to  $J$ . The terminal condition  $J(T, y, x) = 0$  requires  $q_T = h_T = k_T = 0$ . Substituting the trial solution into the partial differential equation gives

$$\begin{aligned} 0 &= \log y(\beta - \beta q_t + q'_t) + x(-d_x - (\beta + \lambda)h_t + s_x q_t + h'_t) \\ &\quad + \bar{x}\lambda h_t + \frac{1}{2}\alpha\eta^2 h_t^2 - \beta k_t + \frac{1}{2}q_t(2\mu - \sigma^2 + \alpha\sigma^2 q_t) + k'_t. \end{aligned}$$

The candidate solution solves the partial differential equation if and only if

$$\begin{aligned} 0 &= \beta - \beta q_t + q'_t, \\ 0 &= -d_x - (\beta + \lambda)h_t + s_x q_t + h'_t, \\ 0 &= \bar{x}\lambda h_t + \frac{1}{2}\alpha\eta^2 h_t^2 - \beta k_t + \frac{1}{2}q_t(2\mu - \sigma^2 + \alpha\sigma^2 q_t) + k'_t, \\ 0 &= q_T = h_T = k_T. \end{aligned}$$

In general, if  $g$  is a continuous function, there is a unique function  $f$  that satisfies the differential equation  $f'_t - af_t + g_t = 0$ , with  $f_T = 0$  and  $a \in \mathbb{R}$ . The solution is

$$f_t = \int_t^T e^{-a(u-t)} g_u du.$$

The formulas for  $q$ ,  $h$  and  $k$  follow directly by applying this result. *q.e.d.*

**Proof of Proposition 2:** An application of Itô's lemma to the expression for  $\pi$  gives the following dynamics

$$\begin{aligned}\frac{d\pi_t}{\pi_t} &= \frac{df_c(\tilde{c}_t, V_t)}{f_c(\tilde{c}_t, V_t)} + f_v(\tilde{c}_t, V_t)dt \\ &= \frac{1}{g(y_t, x_t)} (g_y(y_t, x_t)\mathfrak{D}y_t dt + g_x(y_t, x_t)\mathfrak{D}x_t dt + \frac{1}{2}g_{y,y}(y_t, x_t)d\langle y \rangle_t + \frac{1}{2}g_{x,x}(y_t, x_t)d\langle x \rangle_t) + k(y_t, x_t)dt \\ &\quad + \frac{g_y(y_t, x_t)}{g(y_t, x_t)}(dy_t - \mathfrak{D}y_t dt) + \frac{g_x(y_t, x_t)}{g(y_t, x_t)}(dx_t - \mathfrak{D}x_t dt),\end{aligned}$$

where I set  $g(y_t, x_t) = f_c(\tilde{c}_t, V_t)$  and  $k(y_t, x_t) = f_v(\tilde{c}_t, V_t)$  and  $\mathfrak{D}$  is the infinitesimal drift operator. Since the instantaneous interest rate is defined as the payout rate of a security with price equal to one, we have  $r_t \pi_t dt = -\mathfrak{D}\pi_t dt$ . Therefore,  $r_t$  has affine form:  $r_t = r_0 + r_x x_t$ . The prices of risk obtain as follow:  $-\pi_t(\theta_{y,t}, \theta_{x,t})d\mathbf{B}_t = d\pi_t - \mathfrak{D}\pi_t dt$ . *q.e.d.*

**Proof of Proposition 3 and 4:** Let consider the conditional expectation:

$$\mathbb{E} [\pi_{t+\tau} y_{t+\tau}^\rho \mid \mathcal{F}_t^{y,x}]$$

and guess a trial solution:  $M(t, \pi_t, y_t, x_t) = \pi_t y_t^\rho e^{b_0(\tau) + b_x(\tau)x_t}$ . Then, an application of Itô's lemma to  $M$  gives the following partial differential equation

$$\begin{aligned}0 &= M_t + M_\pi \mathfrak{D}\pi + M_y \mathfrak{D}y + \frac{1}{2}M_{y,y}\langle y \rangle + M_x \mathfrak{D}x + \frac{1}{2}M_{x,x}\langle x \rangle + M_{\pi,y}\langle \pi, y \rangle + M_{\pi,x}\langle \pi, x \rangle \\ &= b'_0(\tau) + b'_x(\tau)x - r_0 - r_x x + \rho(\mu + s_x x) + \rho(\rho - 1)\sigma^2/2 + b_x(\tau)\lambda(\bar{x} - x) + (b_x(\tau)\eta)^2/2 - \theta_y \sigma \rho - \theta_x b_x(\tau)\eta,\end{aligned}$$

with  $b_0(0) = b_x(0) = 0$ . The candidate solution solves the partial differential equation if and only if

$$\begin{aligned}0 &= b'_x(\tau) - r_x + \rho s_x - b_x(\tau)\lambda, \\ 0 &= b'_0(\tau) - r_0 + \rho\mu + \rho(\rho - 1)\sigma^2/2 + b_x(\tau)\lambda\bar{x} + (b_x(\tau)\eta)^2/2 - \theta_y \sigma \rho - \theta_x b_x(\tau)\eta, \\ 0 &= b_0(0) = b_x(0).\end{aligned}$$

Since  $P_{t,\tau} = \pi_t^{-1}M(t, \pi_t, y_t, x_t)$ , once we solve for  $b_0(\tau)$  and  $b_x(\tau)$ , the results of Proposition 4 automatically follow. By setting  $\rho = 0$ , we have  $Q_{t,\tau} = \pi_t^{-1}M(t, \pi_t, y_t, x_t)$  and the results of Proposition 3 follow too. *q.e.d.*

**Proof of Corollary 1 and 2:** The discount bond yield and the equity yield automatically obtain by using the definitions  $\varepsilon(t, \tau) = -\log(Q_{t,\tau})/\tau$  and  $p(t, \tau) = -\log(P_{t,\tau}/\delta_t)/\tau$  and the results of Proposition 3 and 4.

By an application of Itô's lemma to  $Q_{t,\tau}$  and  $P_{t,\tau}$ , we obtain the dynamics:

$$\begin{aligned}\frac{dQ_{t,\tau}}{Q_{t,\tau}} &= (r_t + (\mu_Q - r)(t, \tau))dt + a_x(\tau)\eta dB_{x,t}, \\ \frac{dP_{t,\tau}}{P_{t,\tau}} &= (r_t + (\mu_P - r)(t, \tau))dt + \rho\sigma dB_{y,t} + b_x(\tau)\eta dB_{x,t},\end{aligned}$$

with premia given by

$$\begin{aligned}(\mu_Q - r)(t, \tau)dt &= -\frac{1}{\pi_t Q_{t,\tau}} d\langle \pi, Q \rangle_t, \\(\mu_P - r)(t, \tau)dt &= -\frac{1}{\pi_t P_{t,\tau}} d\langle \pi, P \rangle_t.\end{aligned}$$

*q.e.d.*

## Infinite-horizon, full information economy

**Proposition B** *The process  $V = V(\tilde{c}, \mathbf{F}^{y,x})$  is given by  $V_t = \frac{1}{\alpha} \left( e^{\alpha(q \log y_t + h x_t + k)} - 1 \right)$ , where  $q$ ,  $h$  and  $k$  are deterministic processes, given by*

$$q = 1, \quad h = \frac{s_x - d_x}{\beta + \lambda}, \quad k = \left( \frac{\alpha(s_x - d_x)^2 \eta^2}{(\beta + \lambda)^2} + \frac{2\bar{x}(s_x - d_x)\lambda}{\beta + \lambda} + 2\mu + \sigma^2(\alpha - 1) \right) / (2\beta).$$

**Proof of Proposition B:** An application of Itô's lemma to the expression for  $V$  gives the following partial differential equation for  $V_t = J(y_t, x_t)$ :

$$0 = J_y(\mu + s_x x)y + \frac{1}{2}J_{y,y}\sigma^2 y^2 + J_x \lambda(\bar{x} - x) + \frac{1}{2}J_{x,x}\eta^2 + f(\tilde{c}, J)$$

with  $\tilde{c} = \mathfrak{C}(y, x)$ . A trial solution for  $J$  is  $\log(1 + \alpha J)/\alpha = (q \log y + h x + k)$ . Notice that as long as  $\alpha \rightarrow 0$ , the left hand side converges to  $J$ . Substituting the trial solution into the partial differential equation gives

$$0 = \log y(\beta - \beta q) + x(-d_x - (\beta + \lambda)h + s_x q) + \bar{x}\lambda h + \frac{1}{2}\alpha\eta^2 h^2 - \beta k + \frac{1}{2}q(2\mu - \sigma^2 + \alpha\sigma^2 q).$$

The candidate solution solves the partial differential equation if and only if

$$\begin{aligned}0 &= \beta - \beta q, \\0 &= -d_x - (\beta + \lambda)h + s_x q, \\0 &= \bar{x}\lambda h + \frac{1}{2}\alpha\eta^2 h^2 - \beta k + \frac{1}{2}q(2\mu - \sigma^2 + \alpha\sigma^2 q).\end{aligned}$$

The formulas for  $q$ ,  $h$  and  $k$  follow directly by applying this result. *q.e.d.*

**Proposition C** *The state-price process  $\pi$  satisfies  $d\pi_t = \pi_t(-r_t dt - \theta_y dB_{y,t} - \theta_x dB_{x,t})$ , for any  $t \geq 0$ , where*

$$r_t = \beta + \mu + s_x x_t - \sigma^2(1 - \alpha), \quad \theta_y = \sigma(1 - \alpha), \quad \text{and} \quad \theta_x = \frac{\alpha\eta}{\beta + \lambda}(d_x - s_x).$$

**Proposition D** *The equilibrium price of the equity claim on the dividend  $\delta = y^p$  with residual maturity  $\tau$  is given by  $P_{t,\tau} =$*

$\delta_t e^{b_0(\tau)+b_x(\tau)x_t}, \forall \tau \geq 0$ , where

$$b_0(\tau) = \int_0^\tau \left( \beta - (\rho - 1) (\mu + (\rho - 2)\sigma^2/2) - \eta^2 b_x(u)^2/2 - b_x(u) (\bar{x}\lambda + \alpha\eta^2 h) \right. \\ \left. - \frac{1}{2}\alpha (2\bar{x}\lambda h + \alpha\eta^2 h^2 - 2\beta k + q (2\mu + (2\rho - 3)\sigma^2 + \alpha\sigma^2 q)) \right) du, \\ b_x(\tau) = \frac{(1 - e^{-\tau\lambda}) s_x (\rho - 1)}{\lambda}.$$

The equilibrium price of the discount bond,  $Q_{t,\tau}$ , with residual maturity  $\tau$  obtains as a special case for  $\rho = 0$ .

**Proof of Proposition C and D:** The results of Proposition C and D automatically obtain as a special case of Proposition 2, 3 and 4 for  $T \rightarrow \infty$ : that is by substituting  $q_t, h_t$  and  $k_t$  with the constants  $q, h$  and  $k$  from Proposition B. *q.e.d.*

## Infinite-horizon, incomplete information economy

### Proof of Proposition 5:

i) Let  $y_0, x_0$  and  $s_0$  be random variables with  $x_0$  Gaussian with variance  $\hat{\phi}_0$ , conditional on  $y_0$  and  $s_0$ . By Theorem 12.7 in Lipster and Shirayev (1978), the conditional variance  $\hat{\phi}$  satisfies the following ordinary differential equation:

$$\hat{\phi}'_t = c_0 + c_1 \hat{\phi}_t + c_2 \hat{\phi}_t^2,$$

with

$$c_0 = \eta^2 - (0, \eta v_x) \Sigma^{-1} (0, \eta v_x)', \\ c_1 = -2\lambda - 2(s_x, 1) \Sigma^{-1} (0, \eta v_x)', \\ c_2 = -(s_x, 1) \Sigma^{-1} (s_x, 1)'.$$

The roots are real with  $\hat{\phi}^- < 0 < \hat{\phi}^+$  as long as  $c_0 > 0$  and  $c_2 < 0$ , which is the case under the assumption that  $v'v$  is positive definite. Therefore, the conditional variance  $\hat{\phi}$  has solution

$$\hat{\phi}_t = \hat{\phi}^+ \frac{1 + e^{-\bar{c}t} \frac{\hat{\phi}^-}{\hat{\phi}^+} \left( \frac{\hat{\phi}^+ - \hat{\phi}_0}{\hat{\phi}_0 - \hat{\phi}^-} \right)}{1 + e^{-\bar{c}t} \left( \frac{\hat{\phi}^+ - \hat{\phi}_0}{\hat{\phi}_0 - \hat{\phi}^-} \right)},$$

where  $\hat{\phi}^\pm = -(c_1 \pm \bar{c})/(2c_2)$  and  $\bar{c} = \sqrt{c_1^2 - 4c_0c_2}$ . The steady-state limit of  $\hat{\phi}$  is defined as the limit of  $\hat{\phi}_t$  as  $t \rightarrow \infty$  which is equal to  $\hat{\phi}^+$  for  $\bar{c} > 0$ .

ii) By Theorem 12.7 in Lipster and Shirayev (1978), the process  $\mathbf{W}$  satisfies

$$d\mathbf{W}_t = \begin{pmatrix} dW_{y,t} \\ dW_{s,t} \end{pmatrix} = \Sigma^{-1/2} \begin{pmatrix} dy_t/y_t - (\mu + s_x \hat{x}_t) dt \\ ds_t - \hat{x}_t dt \end{pmatrix}.$$

iii) The result obtains from Lemma 11.3 in [Lipster and Shirayev \(1978\)](#). *q.e.d.*

**Proposition E** Under the dynamics in Eq. (36)-(37)-(8), we have that for any  $t \geq 0$ ,

$$\mathbb{E} [\tilde{c}_{t+\tau}^a | \mathcal{F}_t^o] = y_t^a e^{m_0(\tau) + m_x(\tau)\hat{x}_t}, \quad \tau \geq 0,$$

with

$$\begin{aligned} m_0(\tau) = & -\frac{1}{4\beta^2\lambda^3}a \left( 2\beta\lambda^2(\lambda(\beta(\sigma^2 - 2\mu)\tau - 2(e^{-\lambda\tau} - 1)\bar{x}d_x) - 2\bar{x}\beta(e^{-\lambda\tau} - 1 + \lambda\tau)s_x) \right. \\ & + a \left( \beta^2(3 + e^{-2\lambda\tau} - 4e^{-\lambda\tau} - 2\lambda\tau)s_x^2(\sigma_{xs}^2 + \sigma_{xy}^2) \right. \\ & + \lambda s_x(4\beta^2\sigma(1 - e^{-\lambda\tau} - \lambda\tau)\sigma_{xy} + 2\beta(e^{-\lambda\tau} - 1)^2d_x(\sigma_{xs}^2 + \sigma_{xy}^2)) \\ & \left. \left. + \lambda^2(4(1 - e^{-\lambda\tau})\beta d_x\sigma\sigma_{xy} - 2\beta^2\lambda\sigma^2\tau + (e^{-2\lambda\tau} - 1)d_x^2(\sigma_{xs}^2 + \sigma_{xy}^2)) \right) \right), \\ m_x(\tau) = & a \left( (1 - e^{-\lambda\tau})\frac{s_x}{\lambda} - e^{-\lambda\tau}\frac{d_x}{\beta} \right), \end{aligned}$$

for any  $a \geq 0$  such that the expectation exists finite. Moreover,

$$\frac{\partial_{\hat{x}}\mathbb{E} [\tilde{c}_{t+\tau}^a | \mathcal{F}_t^o]}{\mathbb{E} [\tilde{c}_{t+\tau}^a | \mathcal{F}_t^o]} - \frac{\partial_{\hat{x}}\mathbb{E} [c_{t+\tau}^a | \mathcal{F}_t^o]}{\mathbb{E} [c_{t+\tau}^a | \mathcal{F}_t^o]} \leq 0, \quad \forall d_x \geq 0.$$

**Proof of Proposition E:** Let consider the conditional expectation:

$$\mathbb{E} [\tilde{c}_{t+\tau}^a | \mathcal{F}_t^o], \quad \tau \geq 0,$$

and guess a trial solution:  $M(t, y_t, \hat{x}_t) = y_t^a e^{m_0(\tau) + m_x(\tau)\hat{x}_t}$ . Then, an application of Itô's lemma to  $M$  gives the following partial differential equation

$$\begin{aligned} 0 = & M_t + M_y \mathfrak{D}y + \frac{1}{2}M_{y,y}\langle y \rangle + M_{\hat{x}}\mathfrak{D}\hat{x} + \frac{1}{2}M_{\hat{x},\hat{x}}\langle \hat{x} \rangle + M_{y,\hat{x}}\langle y, \hat{x} \rangle \\ = & m'_0(\tau) + m'_x(\tau)\hat{x} + a(\mu + s_x\hat{x}) + a(a-1)\sigma^2/2 + m_x(\tau)\lambda(\bar{x} - \hat{x}) + m_x(\tau)^2(\sigma_{xy}^2 + \sigma_{xs}^2)/2 + am_x(\tau)\sigma_{xy}\sigma, \end{aligned}$$

with  $m_0(0) = 0$  and  $m_x(0) = -a(d_x/\beta)$  and where  $\mathfrak{D}$  is the infinitesimal drift operator. The candidate solution solves the partial differential equation if and only if

$$\begin{aligned} 0 = & m'_x(\tau) + as_x - m_x(\tau)\lambda, \\ 0 = & m'_0(\tau) + a\mu + a(a-1)\sigma^2/2 + m_x(\tau)\lambda\bar{x} + m_x(\tau)^2(\sigma_{xy}^2 + \sigma_{xs}^2)/2 + am_x(\tau)\sigma_{xy}\sigma, \\ 0 = & m_0(0) = m_x(0) + a(d_x/\beta). \end{aligned}$$

Hence,  $\mathbb{E} [\tilde{c}_{t+\tau}^a | \mathcal{F}_t^o] = M(t, y_t, \hat{x}_t)$ , once we solve for  $m_0(\tau)$  and  $m_x(\tau)$ . By setting  $d_x = 0$ , we have  $\mathbb{E} [c_{t+\tau}^a | \mathcal{F}_t^o] = M(t, y_t, \hat{x}_t)$ . It easily follows that

$$\frac{\partial_{\hat{x}}\mathbb{E} [\tilde{c}_{t+\tau}^a | \mathcal{F}_t^o]}{\mathbb{E} [\tilde{c}_{t+\tau}^a | \mathcal{F}_t^o]} - \frac{\partial_{\hat{x}}\mathbb{E} [c_{t+\tau}^a | \mathcal{F}_t^o]}{\mathbb{E} [c_{t+\tau}^a | \mathcal{F}_t^o]} = -ae^{-\tau}(d_x/\beta) \leq 0,$$

for any  $d_x \geq 0$ . *q.e.d.*

**Proposition F** The process  $V = V(\tilde{c}, \mathbf{F}^{y,s})$  is given by  $V_t = \frac{1}{\alpha} \left( e^{\alpha(q \log y_t + h \hat{x}_t + k)} - 1 \right)$ , where  $q$ ,  $h$  and  $k$  are deterministic processes, given by

$$q = 1, \quad h = \frac{s_x - d_x}{\beta + \lambda}, \quad k = \frac{\mu}{\beta} + \frac{(\alpha - 1)\sigma^2}{2\beta} + \frac{(s_x - d_x)((\sigma_{xs}^2 + \sigma_{xy}^2)\alpha(s_x - d_x) + 2\bar{x}\lambda(\beta + \lambda) + 2\sigma_{xy}\alpha(\beta + \lambda)\sigma)}{2\beta(\beta + \lambda)^2}.$$

**Proof of Proposition F:** An application of Itô's lemma to the expression for  $V$  gives the following partial differential equation for  $V_t = J(y_t, \hat{x}_t)$ :

$$0 = J_y(\mu + s_x \hat{x})y + \frac{1}{2}J_{yy}\sigma^2 y^2 + J_{\hat{x}}\lambda(\bar{x} - \hat{x}) + \frac{1}{2}J_{\hat{x}\hat{x}}(\sigma_{xy}^2 + \sigma_{xs}^2) + J_{y,\hat{x}}\sigma_y\sigma_{xy} + f(\tilde{c}, J)$$

with  $\tilde{c} = \mathfrak{C}(y, \hat{x})$ . A trial solution for  $J$  is  $\log(1 + \alpha J)/\alpha = (q \log y + h \hat{x} + k)$ . Notice that as long as  $\alpha \rightarrow 0$ , the left hand side converges to  $J$ . Substituting the trial solution into the partial differential equation gives

$$0 = \log y(\beta - \beta q) + \hat{x}(-d_x - (\beta + \lambda)h + s_x q) + \bar{x}\lambda h + \frac{1}{2}\alpha(\sigma_{xy}^2 + \sigma_{xs}^2)h^2 - \beta k + \frac{1}{2}q(2\mu - \sigma^2 + \alpha\sigma^2 q) + hq\sigma\sigma_{xy}\alpha.$$

The candidate solution solves the partial differential equation if and only if

$$\begin{aligned} 0 &= \beta - \beta q, \\ 0 &= -d_x - (\beta + \lambda)h + s_x q, \\ 0 &= \bar{x}\lambda h + \frac{1}{2}\alpha(\sigma_{xy}^2 + \sigma_{xs}^2)h^2 - \beta k + \frac{1}{2}q(2\mu - \sigma^2 + \alpha\sigma^2 q) + hq\sigma\sigma_{xy}\alpha. \end{aligned}$$

The formulas for  $q$ ,  $h$  and  $k$  follow directly by applying this result. *q.e.d.*

**Proposition G** The state-price process  $\pi$  satisfies  $d\pi_t = \pi_t(-r_t dt - \theta_y dW_{y,t} - \theta_s dW_{s,t})$ , for any  $t \geq 0$ , where

$$r_t = \beta + \mu + s_x x_t + \sigma \left( \frac{\sigma_{xy}\alpha(s_x - d_x)}{\beta + \lambda} + (\alpha - 1)\sigma \right), \quad \theta_y = \sigma(1 - \alpha) + (d_x - s_x) \frac{\sigma_{xy}\alpha}{\beta + \lambda}, \quad \text{and} \quad \theta_s = (d_x - s_x) \frac{\sigma_{xs}\alpha}{\beta + \lambda}.$$

**Proof of Proposition G:** An application of Itô's lemma to the expression for  $\pi$  gives the following dynamics

$$\begin{aligned} \frac{d\pi_t}{\pi_t} &= \frac{df_c(\tilde{c}_t, V_t)}{f_c(\tilde{c}_t, V_t)} + f_v(\tilde{c}_t, V_t)dt \\ &= \frac{1}{g(y_t, \hat{x}_t)} \left( g_y(y_t, \hat{x}_t)\mathfrak{D}y_t dt + g_{\hat{x}}(y_t, \hat{x}_t)\mathfrak{D}\hat{x}_t dt + \frac{1}{2}g_{yy}(y_t, \hat{x}_t)d\langle y \rangle_t + \frac{1}{2}g_{\hat{x}\hat{x}}(y_t, \hat{x}_t)d\langle \hat{x} \rangle_t + g_{y,\hat{x}}d\langle y, \hat{x} \rangle_t \right) \\ &\quad + k(y_t, \hat{x}_t)dt + \frac{g_y(y_t, \hat{x}_t)}{g(y_t, \hat{x}_t)}(dy_t - \mathfrak{D}y_t dt) + \frac{g_{\hat{x}}(y_t, \hat{x}_t)}{g(y_t, \hat{x}_t)}(d\hat{x}_t - \mathfrak{D}\hat{x}_t dt), \end{aligned}$$

where I set  $g(y_t, \hat{x}_t) = f_c(\tilde{c}_t, V_t)$  and  $k(y_t, \hat{x}_t) = f_v(\tilde{c}_t, V_t)$  and  $\mathfrak{D}$  is the infinitesimal drift operator. Since the instantaneous interest rate is defined as the payout rate of a security with price equal to one, we have  $r_t \pi_t dt = -\mathfrak{D}\pi_t dt$ . Therefore,  $r_t$  has affine form:

$r_t = r_0 + r_x \hat{x}_t$ . The prices of risk obtain as follow:  $-\pi_t(\theta_y, \theta_s)d\mathbf{W}_t = d\pi_t - \mathfrak{D}\pi_t dt$ . *q.e.d.*

**Proposition H** The equilibrium price of the equity claim on the dividend  $\delta = y^p$  with residual maturity  $\tau$  is given by  $P_{t,\tau} = \delta_t e^{b_0(\tau) + b_x(\tau)\hat{x}_t}$ ,  $\forall \tau \geq 0$ , where

$$\begin{aligned} b_0(\tau) &= \int_0^\tau \left( (\mu(\rho - 1) - \beta) + \frac{s_x - d_x}{\beta + \lambda} \alpha(\rho - 1) \sigma \sigma_{xy} + (\rho - 1)((\alpha - 1) + \rho/2) \sigma^2 \right. \\ &\quad \left. + \left( \frac{s_x - d_x}{\beta + \lambda} \alpha(\sigma_{xs}^2 + \sigma_{xy}^2) + \bar{x} \lambda + \sigma_{xy}(\alpha - 1 + \rho) \sigma \right) b_x(u) + b_x(u)^2 (\sigma_{xs}^2 + \sigma_{xy}^2)/2 \right) du, \\ b_x(\tau) &= \frac{(1 - e^{-\tau \lambda}) s_x (\rho - 1)}{\lambda}. \end{aligned}$$

The equilibrium price of the discount bond,  $Q_{t,\tau}$ , with residual maturity  $\tau$  obtains as a special case for  $\rho = 0$ .

**Proof of Proposition H:** Let consider the conditional expectation:

$$\mathbb{E} [\pi_{t+\tau} y_{t+\tau}^\rho \mid \mathcal{F}_t^{y,s}]$$

and guess a trial solution:  $M(t, \pi_t, y_t, \hat{x}_t) = \pi_t y_t^\rho e^{b_0(\tau) + b_x(\tau)\hat{x}_t}$ . Then, an application of Itô's lemma to  $M$  gives the following partial differential equation

$$\begin{aligned} 0 &= M_t + M_\pi \mathfrak{D}\pi + M_y \mathfrak{D}y + \frac{1}{2} M_{y,y} \langle y \rangle + M_{\hat{x}} \mathfrak{D}\hat{x} + \frac{1}{2} M_{\hat{x},\hat{x}} \langle \hat{x} \rangle + M_{\pi,y} \langle \pi, y \rangle + M_{\pi,\hat{x}} \langle \pi, \hat{x} \rangle + M_{y,\hat{x}} \langle y, \hat{x} \rangle \\ &= b'_0(\tau) + b'_x(\tau)x - r_0 - r_x \hat{x} + \rho(\mu + s_x \hat{x}) + \rho(\rho - 1) \sigma^2/2 + b_x(\tau) \lambda (\bar{x} - \hat{x}) + b_x(\tau)^2 (\sigma_{xy}^2 + \sigma_{xs}^2)/2 \\ &\quad - \theta_y (\sigma \rho + b_x(\tau) \sigma_{xy}) - \theta_s b_x(\tau) \sigma_{xs} + b_x(\tau) \sigma \rho \sigma_{xy}, \end{aligned}$$

with  $b_0(0) = b_x(0) = 0$ . The candidate solution solves the partial differential equation if and only if

$$\begin{aligned} 0 &= b'_x(\tau) - r_x + \rho s_x - b_x(\tau) \lambda, \\ 0 &= b'_0(\tau) - r_0 + \rho \mu + \rho(\rho - 1) \sigma^2/2 + b_x(\tau) \lambda \bar{x} + b_x(\tau)^2 (\sigma_{xy}^2 + \sigma_{xs}^2)/2 - \theta_y (\sigma \rho + b_x(\tau) \sigma_{xy}) \\ &\quad - \theta_s b_x(\tau) \sigma_{xs} + b_x(\tau) \sigma \rho \sigma_{xy}, \\ 0 &= b_0(0) = b_x(0). \end{aligned}$$

Since  $P_{t,\tau} = \pi_t^{-1} M(t, \pi_t, y_t, \hat{x}_t)$ , once we solve for  $b_0(\tau)$  and  $b_x(\tau)$ , the equity claim price automatically follows. By setting  $\rho = 0$ , we have  $Q_{t,\tau} = \pi_t^{-1} M(t, \pi_t, y_t, \hat{x}_t)$  and the discount bond price follows too. *q.e.d.*

**Proof of Corollary 3:** The result of Eq. (42) automatically follows from the exact formula for  $V_t$  in Proposition F and the those of the posterior variances in Eq. (39). *q.e.d.*

**Proof of Corollary 4:** The sign of the slopes of the term structures of premia on bond and equity returns are given by

$$\text{sign}(\partial_\tau(\mu_Q - r)(\tau)) = \text{sign}(-\partial_\tau(\mu_Q - r)(\tau)) = \text{sign}\left(\alpha(s_x - d_x) \frac{\sigma_{xy}^2 + \sigma_{xs}^2}{\beta + \lambda} - (1 - \alpha) \sigma \sigma_{xy}\right).$$

Recall that  $\beta + \lambda > 0$ ,  $\sigma \sigma_{xy} > 0$  and  $\alpha < 1$ . Therefore, for  $\alpha(d_x - s_x) \geq 0$   $\partial_\tau(\mu_Q - r)(\tau)$  and  $\partial_\tau(\mu_P - r)(\tau)$  are not equal to zero for any  $v_{ss} > 0$ . Namely, the sign of  $\partial_\tau(\mu_Q - r)(\tau)$  and  $\partial_\tau(\mu_P - r)(\tau)$  are respectively negative and positive for any  $v_{ss} > 0$ . For

$\alpha(d_x - s_x) < 0$ , we have that  $\partial_\tau(\mu_Q - r)(\tau)$  and  $\partial_\tau(\mu_P - r)(\tau)$  are equal to zero for  $v_{ss} = v_{ss}^\pm$  given by

$$v_{ss}^\pm = \frac{\pm(s_x - d_x)\alpha\eta\sigma}{\sqrt{-\alpha^2\eta^2(d_x - s_x)^2s_x^2 + (\alpha - 1)(\beta + \lambda)s_x\sigma^2(2\alpha\lambda d_x - (\beta - \alpha\beta + \lambda + \alpha\lambda)s_x)}}.$$

Since  $v_{ss}^- < 0 < v_{ss}^+$ , I set  $v_{ss}^* = v_{ss}^+$ . Finally, since  $\sigma_{xy}$  is increasing in  $v_{ss}$  and  $\sigma_{xy}^2 + \sigma_{xs}^2$  is decreasing in  $v_{ss}$ , we have that

$$\alpha(s_x - d_x) \frac{\sigma_{xy}^2 + \sigma_{xs}^2}{\beta + \lambda} - (1 - \alpha)\sigma\sigma_{xy}$$

is decreasing in  $v_{ss}$ . Therefore, it automatically obtains that

$$\partial_\tau(\mu_Q - r)(\tau) \geq 0 \quad \text{and} \quad \partial_\tau(\mu_P - r)(\tau) \leq 0 \quad \text{if} \quad v_{ss} \leq v_{ss}^*.$$

By differentiating  $v_{ss}^*$  with respect to  $\gamma = 1 - \alpha$ , we easily find that the sign of  $\partial_\gamma v_{ss}^*$  is given by the sign of the product of  $(d_x - s_x)$  with

$$(\lambda(\alpha d_x - s_x) - s_x\beta(1 - \alpha)).$$

Hence, if  $d_x > s_x$ , then  $\alpha < 0$  and the above term is negative; instead, if  $d_x < s_x$ , then  $0 < \alpha < 1$  and the above term is negative.

Finally we get  $\partial_\gamma v_{ss}^* \geq 0$  if  $\alpha \geq 0$ , given  $\alpha(d_x - s_x) \leq 0$ . *q.e.d.*

**Table I**  
**Baseline model calibration**

| Preferences                              |             |      |
|------------------------------------------|-------------|------|
| time-discount rate                       | $\beta$     | .5%  |
| timing preference                        | $\alpha$    | -9   |
| demand shocks                            | $d_x - s_x$ | 10%  |
| Cash-Flows                               |             |      |
| steady-state of consumption growth       | $\mu$       | 1.5% |
| volatility of consumption growth         | $\sigma$    | 3.5% |
| state process' speed of reversion        | $\lambda$   | .25  |
| state process' volatility                | $\eta$      | 1%   |
| dividends' leverage coefficient          | $\rho$      | 4.5  |
| Implied parameters                       |             |      |
| relative risk aversion                   | $\gamma$    | 10   |
| elasticity of intertemporal substitution | $\psi$      | 1    |
| information quality threshold            | $v_{ss}^*$  | .3%  |

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**Table II**  
**Steady-state moments under full information**

| Data      |  | $r$  | $\sigma_r$ | $\mu_P - r$ | $\sigma_P$ | $SR$ |  |  |  |
|-----------|--|------|------------|-------------|------------|------|--|--|--|
| 1931-2009 |  | .006 | .030       | .062        | .198       | .313 |  |  |  |
| 1947-2009 |  | .010 | .027       | .063        | .176       | .358 |  |  |  |

| Model         |             | $r$   | $\sigma_r$ | $\mu_P - r$ | $\sigma_P$ | $SR$  | $\mu_Q - r$ | (a) | (b) |
|---------------|-------------|-------|------------|-------------|------------|-------|-------------|-----|-----|
| $\alpha = .5$ | $d_x = 0$   | .0194 | .0100      | .0000       | .2107      | .0001 | .0008       | < 0 | > 0 |
| $\alpha = .5$ | $d_x = 1.1$ | .0194 | .0100      | .0030       | .2107      | .0142 | -.0001      | > 0 | < 0 |
| $\alpha = 0$  | $d_x = 0$   | .0188 | .0100      | .0055       | .2107      | .0261 | .0000       | = 0 | = 0 |
| $\alpha = 0$  | $d_x = 1.1$ | .0188 | .0100      | .0055       | .2107      | .0261 | .0000       | = 0 | = 0 |
| $\alpha = -9$ | $d_x = 0$   | .0078 | .0100      | .0998       | .2022      | .4936 | -.0141      | > 0 | < 0 |
| $\alpha = -9$ | $d_x = 1.1$ | .0078 | .0100      | .0502       | .2099      | .2392 | .0014       | < 0 | > 0 |

Steady-state statistics for the risk-free rate ( $r$ ) and its volatility ( $\sigma_r$ ), the excess market return ( $\mu_P - r$ ) and its volatility ( $\sigma_P$ ), the market Sharpe ratio ( $SR$ ) and the excess 30-years bond return ( $\mu_Q - r$ ) are reported for various values of timing preferences ( $\alpha$ ) and demand-effect ( $d_x$ ). Columns (a) and (b) report the sign of the slopes of the term structure of premia respectively on equity and bond returns. Empirical moment statistics are from [Constantinides and Ghosh \(2011\)](#). All unreported parameters are from Table I.

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**Table III**  
**Steady-state moments under incomplete information**

| Data      | $r$  | $\sigma_r$ | $\mu_P - r$ | $\sigma_P$ | $SR$ |
|-----------|------|------------|-------------|------------|------|
| 1931-2009 | .006 | .030       | .062        | .198       | .313 |
| 1947-2009 | .010 | .027       | .063        | .176       | .358 |

Panel A: High information quality ( $v_{ss}^* > v_{ss} = .001$ )

| Model         |             | $r$   | $\sigma_r$ | $\mu_P - r$ | $\sigma_P$ | $SR$  | $\mu_Q - r$ | (a)   | (b)   |
|---------------|-------------|-------|------------|-------------|------------|-------|-------------|-------|-------|
| $\alpha = .5$ | $d_x = 0$   | .0194 | .0098      | .0001       | .2113      | .0005 | .0007       | $< 0$ | $> 0$ |
| $\alpha = .5$ | $d_x = 1.1$ | .0194 | .0098      | .0031       | .2113      | .0147 | -.0001      | $> 0$ | $< 0$ |
| $\alpha = 0$  | $d_x = 0$   | .0188 | .0098      | .0056       | .2113      | .0265 | -.0000      | $> 0$ | $< 0$ |
| $\alpha = 0$  | $d_x = 1.1$ | .0188 | .0098      | .0056       | .2113      | .0265 | -.0000      | $> 0$ | $< 0$ |
| $\alpha = -9$ | $d_x = 0$   | .0074 | .0098      | .1005       | .2030      | .4951 | -.0138      | $> 0$ | $< 0$ |
| $\alpha = -9$ | $d_x = 1.1$ | .0078 | .0098      | .0516       | .2106      | .2450 | .0009       | $< 0$ | $> 0$ |

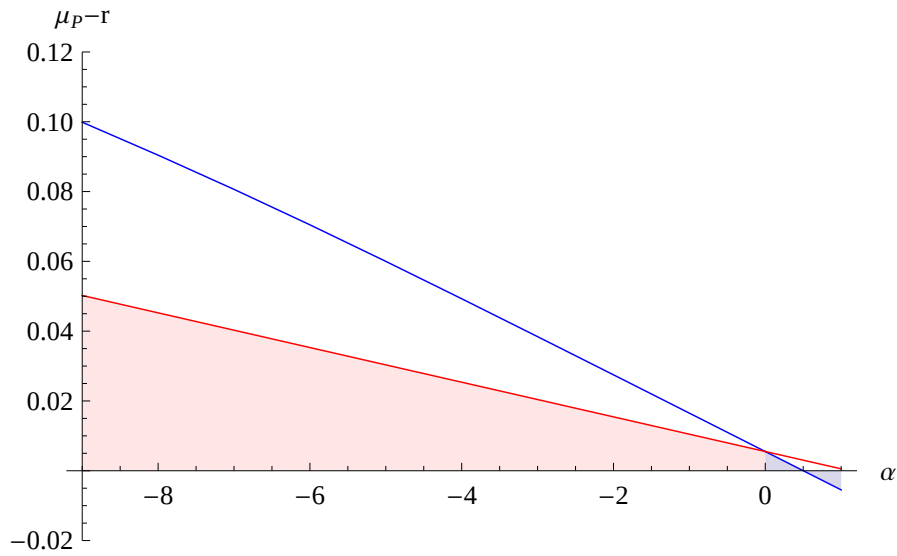
Panel B: Low information quality ( $v_{ss}^* < v_{ss} = .050$ )

| Model         |             | $r$   | $\sigma_r$ | $\mu_P - r$ | $\sigma_P$ | $SR$  | $\mu_Q - r$ | (a)   | (b)   |
|---------------|-------------|-------|------------|-------------|------------|-------|-------------|-------|-------|
| $\alpha = .5$ | $d_x = 0$   | .0197 | .0051      | .0018       | .2202      | .0082 | -.0001      | $> 0$ | $< 0$ |
| $\alpha = .5$ | $d_x = 1.1$ | .0194 | .0051      | .0040       | .2202      | .0182 | -.0003      | $> 0$ | $< 0$ |
| $\alpha = 0$  | $d_x = 0$   | .0188 | .0051      | .0076       | .2202      | .0345 | -.0006      | $> 0$ | $< 0$ |
| $\alpha = 0$  | $d_x = 1.1$ | .0188 | .0051      | .0076       | .2202      | .0345 | -.0006      | $> 0$ | $< 0$ |
| $\alpha = -9$ | $d_x = 0$   | .0025 | .0051      | .1089       | .2139      | .5091 | -.0096      | $> 0$ | $< 0$ |
| $\alpha = -9$ | $d_x = 1.1$ | .0083 | .0051      | .0716       | .2186      | .3275 | -.0055      | $> 0$ | $< 0$ |

Steady-state statistics for the risk-free rate ( $r$ ) and its volatility ( $\sigma_r$ ), the excess market return ( $\mu_P - r$ ) and its volatility ( $\sigma_P$ ), the market Sharpe ratio ( $SR$ ) and the excess 30-years bond return ( $\mu_Q - r$ ) are reported for various values of timing preferences ( $\alpha$ ) and demand-effect ( $d_x$ ). Columns (a) and (b) report the sign of the slopes of the term structure of premia respectively on equity and bond returns. Panels A and B consider the cases  $v_{ss} = .001$  and  $v_{ss} = .05$ , which are respectively below and above the information quality threshold  $v_{ss}^*$ , as computed in the cases of interest (that is, for  $\alpha(d_x - s_x) < 0$ :  $v_{ss} = .03$  for  $\alpha = .5$  and  $d_x = 0$  and  $v_{ss} = .003$  for  $\alpha = -9$  and  $d_x = 1.1$ ). Empirical moment statistics are from [Constantinides and Ghosh \(2011\)](#). All unreported parameters are from Table I.

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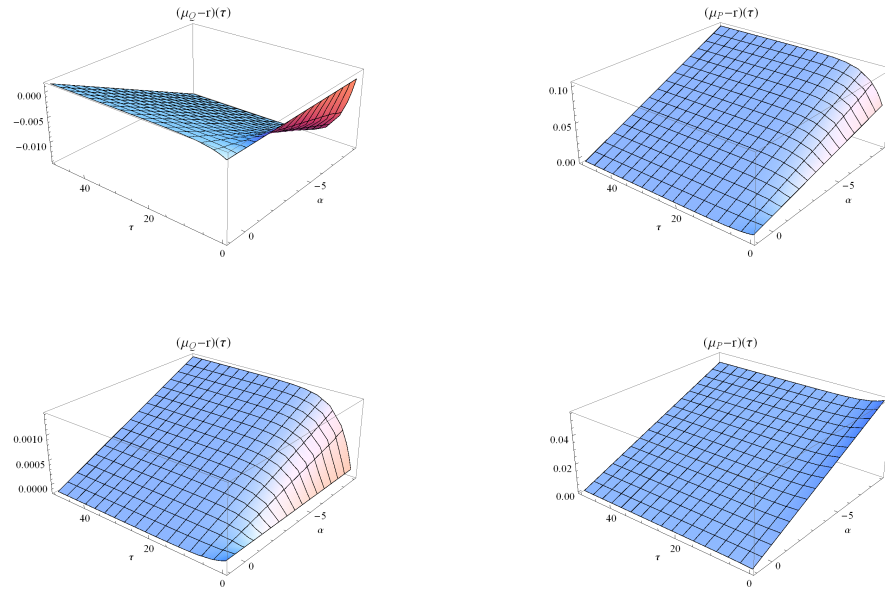
**Figure 1.** Equity premium and timing preferences



The equilibrium equity premium,  $(\mu_P - r)(t)$ , as a function of the timing preferences,  $\alpha$ , for  $d_x = 0$  and  $d_x = 1.1$  is plotted respectively with colours blue and red. The shadow areas denote the cases in which the model implies positive and negative slopes to the term-structures of premia respectively on bond and equity returns. All parameters are from Table I.

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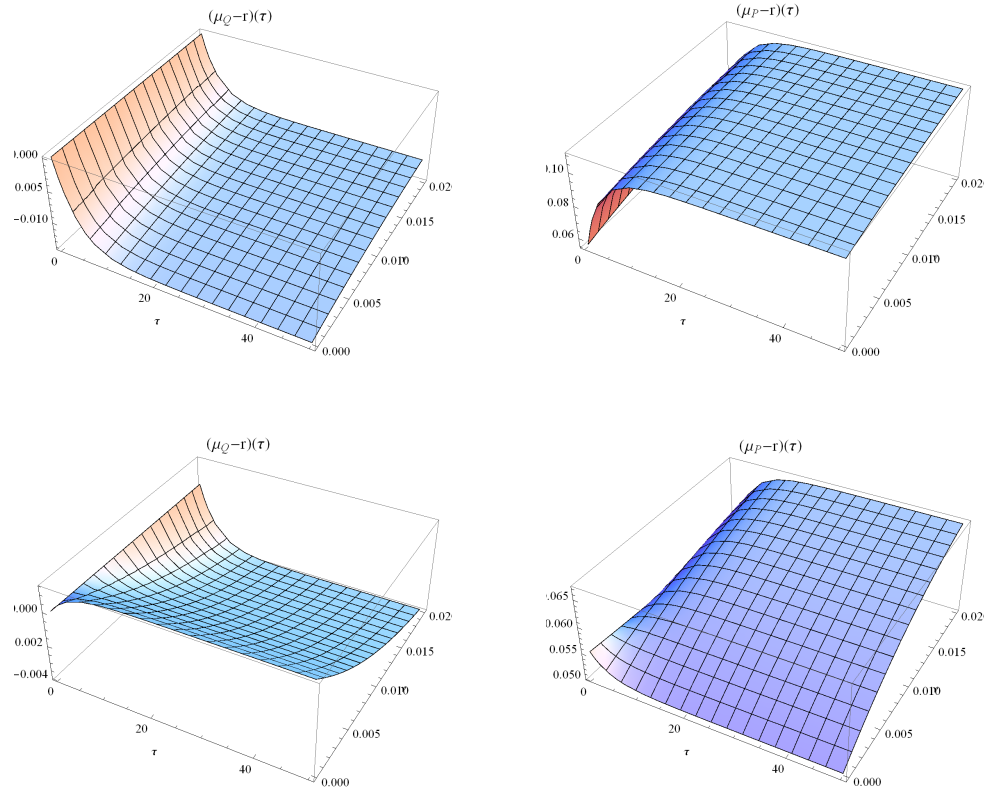
**Figure 2.** Term structures of bond and equity premia and timing preferences



Term structures of premia on bond (left panels) and equity (right panels) premia as a function of the maturity ( $\tau$ ) and the timing preferences ( $\alpha$ ), in the cases  $d_x = 0$  (upper panels) and  $d_x = 1.1$  (lower panels). All parameters are from Table I.

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**Figure 3.** Term structures of bond and equity premia and information quality



Term structures of premia on bond (left panels) and equity (right panels) premia as a function of the maturity ( $\tau$ ) and the information quality ( $v_{ss}$ ), in the cases  $d_x = 0$  (upper panels) and  $d_x = 1.1$  (lower panels). All parameters are from Table 1.

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